

JS Office Suite

MVP Design Document + Feature Parity Checklist

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MVP Scope: Offline-first, localStorage persistence, JSON import/export. Cloud sync is a post-MVP milestone.

Codex Prompt (Paste into your coding agent)

You are implementing a JavaScript-only, offline-first Office suite (Writer, Sheets, Slides, Base).

Below is a design document describing UX, architecture, data models, and feature parity targets.

Scroll to the bottom of the document for the master task checklist.

Rule: treat every checklist item as REQUIRED unless explicitly tagged "Post-MVP". Complete every task for each product area (Shared Shell, Writer, Sheets, Slides, Base).

When implementing a feature:

- 1) Read the corresponding design section for intent, UX behavior, and edge cases.
- 2) Implement the feature end-to-end (UI, model, persistence, undo/redo, accessibility, tests).
- 3) Mark the checklist item as completed by changing ☒ to ☐ in the task list and recording a short verification note.

Constraints:

- JavaScript-only client app. No server required for MVP.
- Persistence: browser localStorage for MVP. Also support JSON export/import for backup/transfer.
- Desktop + mobile UX must be first-class (responsive layouts, touch interactions, keyboard shortcuts).
- Do not assume Office file-format compatibility (DOCX/XLSX/PPTX/ACCDB) in MVP; target parity in behavior and user workflows first.

Quality bar:

- Fast and stable on mid-range phones and laptops.
- Accessible (keyboard + screen reader), and resilient (autosave, crash recovery).
- No data loss: every edit must be persisted or recoverable.

Table of Contents (update in Word)

This document uses heading styles. In Word: References -> Table of Contents -> Automatic Table to generate.

Sections:

- 1. Vision, Principles, Scope
- 2. Shared Suite Shell + Storage
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- 8. Testing, Performance, Accessibility
- 9. Implementation Plan
- 10. Master Task Checklist (CHECKBOXES)
- Appendices: JSON Schemas, Shortcuts, References

1. Vision, Principles, and Scope

Goal: deliver a practical, browser-first Office suite that feels familiar to Microsoft Office users, but is simpler, faster, and mobile-friendly. The suite is a single web app (PWA optional) with four apps:

- Writer: word processor (Word-like)
- Sheets: spreadsheet (Excel-like)
- Slides: presentations (PowerPoint-like)
- Base: small relational database builder (Access-like)

MVP definition: offline-first, local-only, single-user, with robust editing, undo/redo, and export/import via JSON. Cloud sync, collaboration, and Office file-format compatibility are post-MVP.

1.1 Non-goals (MVP)

- Real-time multi-user collaboration (CRDT/OT) and presence indicators
- Native Office file compatibility (DOCX/XLSX/PPTX/ACCDB) - can be added later via import/export pipelines
- Macros/VBA parity - consider a safe scripting sandbox post-MVP
- Full printing/layout parity with desktop Office in every edge case (MVP targets 'good enough' with predictable output)
- Enterprise features (IRM, DLP, admin policy controls)

1.2 Design principles

- Offline-first: the app works with no network and never blocks on remote resources.
- No data loss: autosave, version snapshots, and recovery are mandatory.
- Familiar but not bloated: keep Office mental models (ribbon-ish commands, panes, status bar) but simplify.
- Mobile-first interactions: touch handles, context toolbars, and thumbs-friendly navigation.
- Accessibility is a feature: full keyboard navigation and screen-reader semantics.
- Performance: handle large docs/spreadsheets with virtualization and incremental rendering.
- Extensibility: a shared command system so new tools can plug into menus, shortcuts, and the command palette.

2. Shared Suite Shell + Storage

2.1 App shell (one SPA, four workspaces)

The suite is a single-page app with a shared shell: document library, templates, search, settings, and app switcher. Each product loads as a workspace inside the shell, sharing commands, theming, storage, and export/import.

2.2 Navigation model

Desktop layout:

```
+-----+
--+
| Top Bar: [App Switcher]  Search...           [Share*] [Export] [Settings]
|
+-----+
+-----+
| Left Rail                | Editor Surface
|
| - Writer                 | - Context Ribbon (tabs)
|
| - Sheets                 | - Canvas/DOM editor
|
| - Slides                 | - Right Inspector (optional)
|
| - Base                   |
|
+-----+
+-----+
| Status Bar: mode / selection / zoom / messages | autosave indicator |
+-----+
--+
```

* Share is post-MVP (cloud). MVP shows 'Export/Import'.

Mobile layout (thumb-first):

```
+-----+
| Top: Document name ▼    Undo  Redo  ...      |
+-----+
| Context Toolbar (scrollable chips):          |
| [B] [I] [U] [Styles] [Insert] [Table] ...   |
+-----+
| Editor Surface (gesture + handles)           |
|                                               |
+-----+
| Bottom Tab Bar: Library | Writer | Sheets |  |
| Slides | Base | (or: app icons + recent)    |
+-----+
```

2.3 Storage (MVP)

All data is stored in browser localStorage. The suite maintains a workspace index (metadata) and stores each document payload as compressed JSON strings (or chunked strings) to avoid size limits. If localStorage limits are hit, surface a clear error and offer immediate JSON export as a fallback.

2.4 Import/Export

Every document can be exported as JSON. Import creates a new document (or updates an existing one if IDs match). Also provide bulk export/import (entire workspace) for backups.

2.5 Autosave + recovery

Autosave runs after every change batch (debounced). Keep a small rolling history of snapshots (e.g., last 20) per document. On load, detect corrupted payloads and offer recovery from the most recent valid snapshot.

3. UI System (Desktop + Mobile)

3.1 Command system (the spine of the suite)

All actions are commands: they have IDs, labels, icons, shortcuts, enable/disable rules, and handlers. Menus, ribbon tabs, context menus, and the command palette all render from the same command registry.

3.2 Ribbon + contextual toolbars

Desktop uses a ribbon-inspired layout with fewer tabs: Home, Insert, Layout, Review, View, plus app-specific tabs. Mobile uses a horizontal chip toolbar that swaps based on selection.

3.3 Inspector pane

A right-side inspector shows properties for the selected object (text, shape, cell range, slide object, field). On mobile it appears as a bottom sheet.

3.4 Accessibility

- Full keyboard navigation for every interactive control (Tab/Shift-Tab, arrow key roving, Enter/Space).
- ARIA roles for menus, toolbars, dialogs, grids, tree views, and property panels.
- High-contrast mode and reduced-motion mode.
- Screen-reader friendly selection announcements (e.g., 'Cell B2 selected', 'Paragraph style: Heading 2').

3.5 Touch interactions

- Selection handles for text ranges, table cells, shapes, and slide objects.
- Two-finger scroll/pan without accidental selections.
- Pinch-to-zoom (Sheets and Slides), and double-tap to edit.
- Long-press opens context menu and shows command chips near finger.

4. Writer (Word-like) Design

Writer is a structured document editor: it edits a document model (not raw HTML) and renders to the DOM. Avoid storing content as arbitrary HTML; store as a clean JSON schema that can evolve.

4.1 Primary workflows

- Create: pick a template -> start typing -> autosave.
- Format: apply styles, paragraphs, lists; quick formatting via toolbar or shortcuts.
- Insert: tables, images, links, horizontal rules, page breaks.
- Review: comments (MVP), track changes (post-MVP), compare/merge (post-MVP).
- Export: JSON (MVP), PDF/print (MVP basic), DOCX (post-MVP).

4.2 Document model (JSON)

```
{
  "id": "doc_...",
  "type": "writer",
  "meta": { "title": "...", "createdAt": "...", "updatedAt": "...", "page":
{ "size": "Letter", "margins": { "t": 72, "r": 72, "b": 72, "l": 72 } } },
  "content": [
    { "block": "paragraph", "style": "Normal", "runs": [ { "text": "Hello",
"marks": ["bold"] } ] },
    { "block": "heading", "level": 1, "runs": [ { "text": "Title" } ] },
    { "block": "table", "rows": [ [ { "runs": [ { "text": "A1" } ] } ] ] }
  ],
  "comments": [ /* thread model */ ],
  "history": { /* snapshot pointers */ }
}
```

4.3 Layout + pagination

MVP uses a paged editing surface (WYSIWYG-ish) with visible page boundaries. Rendering is incremental: only visible pages are laid out. Print/PDF uses the same layout engine to avoid surprises.

4.4 UI specifics

Desktop ribbons for Writer:

- Home: clipboard, font, paragraph, styles, find/replace
- Insert: image, table, link, page break, header/footer
- Layout: margins, orientation, columns, page color
- Review: spelling, comments, (post-MVP: track changes, compare/merge)
- View: zoom, rulers, outline mode

Mobile: show context chips based on cursor/selection: text, paragraph, object (image/table).

4.5 MVP feature set

- Rich text editing: bold/italic/underline/strike, font family/size, color, highlight
- Paragraph: alignment, line spacing, indentation, bullets/numbering, multilevel lists (basic)

- Styles: Normal + Heading 1-3 + Title + Subtitle + Quote + Code; custom styles post-MVP
- Tables: create, add/remove rows/cols, merge/split cells (basic), cell shading and borders (basic)
- Images: insert (paste/drag), resize, wrap (inline + simple wrap), alt text
- Links: insert/edit, open in new tab
- Headers/footers: simple per-doc header/footer text + page number placeholder
- Find/replace: plain text, case-sensitive toggle
- Spelling/grammar: basic spellcheck via browser + custom word list
- Comments: anchored comment threads on text selections
- Export: JSON; Print to PDF via browser print; basic doc statistics (word count)

4.6 Post-MVP parity targets

- Track changes (insert/delete/format change marks) and accept/reject flows
- Compare and merge documents (two versions) into a combined revision view
- Footnotes/endnotes, citations manager, and table of contents generation
- Advanced tables (cell vertical alignment, auto-fit, repeat header row)
- Advanced layout: sections, columns, page breaks with section settings
- Mail merge-like template variables
- DOCX import/export

4.7 Edge cases + acceptance criteria (Writer)

- Undo/redo: every edit operation (typing, formatting, insert object) is reversible without corruption.
- Clipboard: copy/paste preserves styles inside Writer; sanitizes external HTML into safe structured content.
- Performance: typing latency stays under 16ms average on mid-range laptop for 30-page doc.
- Persistence: after a hard refresh, the last autosaved state loads with no more than 5 seconds of lost edits.
- Accessibility: all editing + formatting is usable without a mouse; selection and formatting state announced to screen readers.

5. Sheets (Excel-like) Design

Sheets is a high-performance grid with a formula engine, formatting, and data tools. Rendering uses virtualization: only visible cells are in the DOM/canvas.

5.1 Workbook model (JSON)

```
{
  "id": "wb_...",
  "type": "sheets",
  "meta": { "title": "...", "createdAt": "...", "updatedAt": "..." },
  "sheets": [
    {
      "name": "Sheet1",
      "grid": { "rows": 1000, "cols": 26 },
      "cells": { "A1": { "v": 123, "t": "n" }, "B1": { "f": "=A1*2", "t": "f" } },
      "formats": { "A1": { "num": "0.00", "font": { "b": true } } },
      "colWidths": { "A": 96 },
      "rowHeights": { "1": 20 },
      "namedRanges": [ { "name": "Sales", "ref": "Sheet1!A1:D20" } ],
      "tables": [ /* structured tables */ ],
      "filters": [ /* filter state */ ],
      "conditionalFormats": [ /* rules */ ],
      "dataValidations": [ /* rules */ ]
    }
  ],
  "calc": { "mode": "auto", "locale": "en-US" }
}
```

5.2 Grid UI

Desktop:

```
+-----+-----+
--+
| Formula Bar: fx  [ =SUM(A1:A10) ]                               Name Box [A1]
|
+-----+-----+
+---+
|      | Column headers                                           |
|      |
| Row# | [virtualized cell grid]                                   |
ins|
|      | - selection outline                                       |
pec|
|      | - fill handle                                             |
tor|
+-----+-----+
+---+
```

| Sheet Tabs: [Sheet1] [+] | Status: Ready | Zoom 100% | Calc |
+-----
--+

Mobile:

- - Swipe sheet tabs; tap to edit cell; formula bar becomes a full-width editor.
- - Bottom chips: Format | Insert | Data | Chart | fx | More
- - Drag handles for range selection; long-press for context menu.

5.3 Calculation engine

Implement a formula parser -> AST -> dependency graph. Recalc is incremental: change a cell, re-evaluate only dependents. Support circular reference detection and show helpful errors (#VALUE!, #DIV/0!, #REF!, #NAME?, #N/A).

MVP function set (minimum useful subset):

- Math: SUM, AVERAGE, MIN, MAX, COUNT, COUNTA, ROUND, ABS
- Logical: IF, AND, OR, NOT
- Text: CONCAT, LEFT, RIGHT, MID, LEN, TRIM, LOWER, UPPER
- Lookup: VLOOKUP (basic), XLOOKUP (post-MVP), INDEX+MATCH (post-MVP)
- Date/time: TODAY, NOW, DATE, YEAR, MONTH, DAY
- Ranges: relative/absolute references (\$A\$1), ranges (A1:B10), cross-sheet refs (Sheet2!A1)

5.4 MVP feature set

- Cell editing: value types (number, text, date), inline edit, multi-cell paste, autofill (basic)
- Formatting: font, size, color, fill, borders, number formats, alignment, wrap text
- Row/column ops: insert/delete, resize, hide/show, freeze panes (top row/left column + arbitrary)
- Data tools: sort, filter, find/replace, remove duplicates (basic)
- Data validation: list dropdown, numeric/date/time constraints, input messages (MVP subset)
- Conditional formatting: value-based colors, data bars, icon sets (post-MVP full)
- Charts: bar/line/pie/scatter (basic), edit chart title/legend/series
- Multiple sheets: add/rename/duplicate/move, sheet protection (post-MVP)
- Import/export: CSV import/export per sheet; JSON for workbook

5.5 Post-MVP parity targets

- PivotTables and PivotCharts (fields area, grouping, sorting, filtering)
- Advanced conditional formatting (formula-based rules, icon sets, top/bottom rules)
- Advanced data validation (custom formulas; dependent dropdowns)
- Tables (structured references), slicers, timeline controls
- Power Query-like import (post-MVP), external connectors
- Rich charting: combo charts, axis scaling, trendlines
- Compatibility import/export for XLSX (post-MVP)

5.6 Edge cases + acceptance criteria (Sheets)

- Grid performance: scrolling stays smooth (≥ 45 fps) for 100k populated cells with virtualization.
- Calc correctness: dependency graph updates are correct; volatile functions (TODAY/NOW) update on schedule.
- Copy/paste: preserves formulas and relative references correctly when pasting into different ranges.
- Formatting: applying formats to a range is atomic and undoable.
- Persistence: reopening a workbook restores grid, formats, validations, and named ranges exactly.

6. Slides (PowerPoint-like) Design

Slides is an object-based presentation editor: slides contain shapes, text boxes, images, and charts. Rendering uses SVG/Canvas (pick one) with a property inspector and drag handles.

6.1 Presentation model (JSON)

```
{
  "id": "pres_...",
  "type": "slides",
  "meta": { "title": "...", "createdAt": "...", "updatedAt": "...",
    "aspect": "16:9" },
  "theme": { "name": "Default", "colors": {...}, "fonts": {...} },
  "masters": [ { "id": "m1", "layouts": [ { "id": "l_title", "placeholders": [...] } ]
  } ],
  "slides": [
    { "id": "s1", "layout": "l_title", "objects": [ { "id": "t1", "type": "textbox",
      "x": 120, "y": 80, "w": 720, "h": 120, "text": {...} } ],
      "transition": { "type": "fade", "durationMs": 600 },
      "notes": "Speaker notes..."
    }
  ]
}
```

6.2 Editor UI

- Left: slide thumbnails (virtualized). Center: canvas. Right: inspector.
- Top: context ribbon (Home, Insert, Design, Transitions, Animations, Slide Show, View).
- Mobile: thumbnails become a drawer; inspector is a bottom sheet.

6.3 MVP feature set

- Slides: add/duplicate/delete; reorder via drag; multi-select slides
- Layouts: title slide, title+content, two-column, blank; apply layout; reset to layout
- Text boxes: edit text; basic typography; align/line spacing; bullets/numbering (basic)
- Shapes: rectangle/ellipse/line/arrow; fill/stroke; corner radius (optional)
- Images: insert, crop (basic), mask (post-MVP)
- Alignment: snap guides, align/distribute, group/ungroup, arrange (front/back)
- Themes: choose from presets; basic color/font sets
- Transitions: none, fade, push, wipe; duration control
- Animations: appear/fade for objects (basic), order list (basic)
- Presenter mode: full-screen slide show; arrow navigation; notes view (basic)
- Export: JSON; export to PDF (render slides to images then pack into PDF) (MVP basic)

6.4 Post-MVP parity targets

- Slide Master editing: edit masters and layouts, placeholders, theme fonts/colors
- Advanced transitions (Morph-like), sound triggers (optional)
- Advanced animations: motion paths, triggers, timing, build sequences

- Charts and tables as native objects (shared chart engine from Sheets)
- Import/export PPTX (post-MVP)
- Advanced presenter view: timer, next slide preview, laser pointer

6.5 Edge cases + acceptance criteria (Slides)

- Object manipulation: drag, resize, rotate is precise; snapping is predictable; undo restores positions.
- Text editing: caret behavior consistent; auto-size text boxes (optional) is stable.
- Slide show: no jank when advancing; transitions respect reduced-motion settings.
- Export: PDF pages match slide aspect and safe margins; fonts embedded or substituted predictably.
- Persistence: theme, slide order, and all object properties restore accurately.

7. Base (Access-like) Design

Base is a lightweight relational database builder for small projects. It combines a schema designer, a datasheet editor, a query builder, and simple forms/reports. The entire database is stored as JSON in localStorage.

7.1 Database model (JSON)

```
{
  "id": "db_...",
  "type": "base",
  "meta": { "title": "...", "createdAt": "...", "updatedAt": "..." },
  "tables": [
    { "id": "t_customers", "name": "Customers",
      "fields": [
        { "id": "f_id", "name": "ID", "type": "autonumber", "pk": true },
        { "id": "f_name", "name": "Name", "type": "text", "required": true },
        { "id": "f_email", "name": "Email", "type": "text", "unique": true }
      ],
      "rows": [ { "f_id": 1, "f_name": "Ada", "f_email": "ada@example.com" } ]
    }
  ],
  "relationships": [ { "from": "Orders.customerId", "to": "Customers.ID",
    "onDelete": "restrict" } ],
  "queries": [ { "id": "q1", "name": "ActiveCustomers", "sql": "SELECT ...", "ui":
    {...} } ],
  "forms": [ { "id": "form1", "table": "Customers", "layout": {...} } ],
  "reports": [ { "id": "rep1", "query": "q1", "layout": {...} } ]
}
```

7.2 UI structure

- Left navigation tree: Tables, Queries, Forms, Reports.
- Center: tabbed documents (datasheet, designer, builder).
- Right inspector: field properties, validation rules, formatting.
- Mobile: navigation tree becomes a drawer; designers use step-by-step wizards.

7.3 MVP feature set

- Table designer: create tables and fields; field types (text, number, date, boolean, autonumber, choice)
- Datasheet view: edit rows, sort, filter, column resize, inline validation errors
- Simple relationships: one-to-many with foreign keys; enforce referential integrity (basic)
- Query builder: select fields, filters, sort, simple joins (MVP subset); run query to view results
- Forms: auto-generate a form from a table/query; basic layout (label + control); validation messages
- Reports: simple list report (table) with grouping (basic), export to PDF/print
- Import/export: JSON; CSV import/export per table

7.4 Post-MVP parity targets

- Advanced query designer: aggregate/group by, calculated fields, parameters, saved queries
- Form designer: drag/drop controls, subforms, lookup fields, conditional visibility
- Report designer: headers/footers, page numbers, grouped reports, charts
- Macros/automations (safe sandbox), button actions, navigation
- Role-based security and sharing (requires backend) (post-MVP)
- ODBC/connectors/external databases (post-MVP)

7.5 Edge cases + acceptance criteria (Base)

- Schema changes: adding/removing fields migrates existing rows safely and is undoable (or versioned).
- Validation: required/unique/choice constraints enforced on edit with clear messages.
- Query correctness: joins and filters produce stable, deterministic results; errors are explainable.
- Performance: handles 50k rows across tables with pagination/virtualization in datasheet.
- Export/import: round-trips without losing schema, relationships, or saved queries.

8. Testing, Performance, Security, Accessibility

8.1 Testing strategy

- Unit tests: command handlers, parsers (formulas), model transforms, serialization.
- Integration tests: editor interactions (typing, selections), undo/redo, clipboard, persistence.
- Golden tests: JSON export/import round-trip; rendering snapshots for key documents.
- Performance tests: large docs, large sheets, large slide decks; measure input latency and memory.
- Accessibility audits: keyboard-only flows and screen-reader smoke tests.

8.2 Performance guidelines

- Virtualize everything big: pages, grid rows/cols, slide thumbnails, database rows.
- Debounce autosave and expensive layout; keep editing responsive.
- Use requestAnimationFrame for rendering and pointer interactions.
- Avoid layout thrash: batch DOM reads/writes; use transforms for drag/resize.
- Store derived data (indexes, calc results) separately and recompute incrementally.

8.3 Security model (client-only)

- Sanitize all pasted HTML. Never execute embedded scripts.
- Treat imported JSON as untrusted: validate schema, size limits, and reject dangerous payloads.
- Use Content Security Policy (CSP) and avoid eval/new Function.
- If adding plugins later, sandbox them (iframe + postMessage) and use explicit permissions.

9. Implementation Plan (MVP to Post-MVP)

This plan prioritizes the shared foundation first, then vertical slices for each app. Avoid building four separate apps; build shared command system, persistence, and UI primitives once.

9.1 Milestones

- M0: Project skeleton, command registry, routing, theme system, storage adapter (localStorage).
- M1: Document library + templates + JSON export/import + autosave + snapshot recovery.
- M2: Writer vertical slice (basic rich text, styles, tables, images, comments).
- M3: Sheets vertical slice (grid virtualization, formula engine MVP, formatting, charts basic).
- M4: Slides vertical slice (objects, themes, transitions MVP, PDF export basic).
- M5: Base vertical slice (tables + datasheet + query builder MVP, forms/reports basic).
- M6: Hardening: performance, accessibility, cross-browser testing, edge-case fixes.
- Post-MVP: cloud sync, collaboration, Office format compatibility pipelines.

9.2 Definition of Done (for every task)

- Implemented UI + model + persistence (localStorage) + undo/redo.
- Works on desktop (mouse + keyboard) and mobile (touch) with responsive layout.
- Has error handling and user feedback (toasts/status).
- Covered by tests appropriate to the component (unit/integration).
- Documented briefly in this doc (notes field) or in a /docs folder in the repo.

Appendix A. JSON Schemas (human-readable)

These schemas are descriptive, not strict JSON Schema files. MVP: validate shape + required fields; post-MVP: maintain formal JSON Schema and migration scripts.

A.1 Shared Document Metadata

```
{
  "id": "string (stable UUID)",
  "type": "writer | sheets | slides | base",
  "title": "string",
  "createdAt": "ISO-8601",
  "updatedAt": "ISO-8601",
  "thumbnail": "optional data URL",
  "version": "integer schema version",
  "tags": ["string"],
  "pinned": "boolean"
}
```

A.2 Writer Content Nodes

- - paragraph: runs[]; paragraphProps (align, indent, spacing)
- - heading: level 1-6; runs[]
- - list: ordered/unordered; items[] each with blocks
- - table: rows[]; cell blocks[]
- - image: src, w, h, alt, wrapMode
- - break: pageBreak/sectionBreak
- - commentAnchor: anchorId, from/to positions

A.3 Sheets Cell Encodings

Cell object:

```
{
  "t": "n | s | b | d | e | f (number/string/bool/date/error/formula)",
  "v": "raw value (if not formula)",
  "f": "formula string starting with =",
  "r": "cached result (optional) for faster open",
  "fmt": "format id (optional)",
  "note": "cell note/comment (optional)"
}
```

A.4 Slides Object Types

- - textbox: x, y, w, h, rotation, textRuns, paragraphProps
- - shape: kind, x, y, w, h, fill, stroke, cornerRadius
- - image: src, crop, x, y, w, h
- - group: children[], transform
- - chart (post-MVP): datasetRef, options

A.5 Base Schema

- - table: fields[], rows[]; constraints; indexes
- - relationship: fromTable/fromField -> toTable/toField with onDelete behavior
- - query: uiBuilderState + compiled AST (preferred over raw SQL for safety)
- - form: layout tree of controls bound to fields
- - report: layout + grouping + computed fields

Appendix B. Keyboard Shortcuts (proposed defaults)

Shortcuts should feel familiar, but avoid breaking browser/system defaults. On macOS use Cmd; on Windows/Linux use Ctrl.

B.1 Global

- Ctrl/Cmd+K: Command palette
- Ctrl/Cmd+S: Save (shows toast; autosave still runs)
- Ctrl/Cmd+Z / Shift+Ctrl/Cmd+Z: Undo / Redo
- Ctrl/Cmd+F: Find
- Ctrl/Cmd+P: Print (Writer/Slides/Reports)

B.2 Writer

- Ctrl/Cmd+B/I/U: Bold/Italic/Underline
- Ctrl/Cmd+L/E/R/J: Align left/center/right/justify
- Ctrl/Cmd+Shift+L: Toggle bullets
- Ctrl/Cmd+Alt+1/2/3: Heading 1/2/3
- Ctrl/Cmd+Shift+H: Highlight (cycle)
- Ctrl/Cmd+Shift+\: Clear formatting (post-MVP)

B.3 Sheets

- Enter: commit + move down; Shift+Enter: move up
- Tab: commit + move right; Shift+Tab: move left
- F2: edit cell (desktop)
- Ctrl/Cmd+Arrow: jump to edge of data region (post-MVP)
- Ctrl/Cmd+Space: select column; Shift+Space: select row (post-MVP)

B.4 Slides

- Ctrl/Cmd+M: New slide
- Ctrl/Cmd+D: Duplicate selected object/slide
- Arrow keys: nudge object (Shift for larger step)
- F5: Start slideshow (desktop) (post-MVP key)

B.5 Base

- Ctrl/Cmd+N: New table/query/form/report (opens create menu)
- Ctrl/Cmd+Enter: Open selected item
- Ctrl/Cmd+F: Find in datasheet

Appendix C. UX Copy + Error Messages (starter list)

Keep messages short, specific, and action-oriented. Always offer a safe fallback (export) when storage is at risk.

Key	Message	Notes
Autosave	Saved	Displayed after successful autosave batch.
StorageFull	Storage is full. Export now to avoid losing work.	Show storage usage and offer Export JSON.
ImportInvalid	That file doesn't look like a valid document.	Show validation errors and expected types.
RecoverFound	We found an unsaved session. Restore it?	Provide Restore / Discard options.
CorruptDoc	This document looks corrupted. Restore an earlier snapshot?	Show snapshot list with timestamps.
UndoEmpty	Nothing to undo	Optional toast; avoid spamming.

Appendix D. References (for parity research)

These references informed feature parity targets and terminology. Accessed January 2026.

- Microsoft Support: Word features comparison (web vs desktop).
- Microsoft Support: Basic tasks in Word (including Track Changes).
- Microsoft Support: Compare and merge two versions of a document.
- Microsoft Support: PivotTables and PivotCharts overview; Create a PivotTable.
- Microsoft Support: Use conditional formatting in Excel.
- Microsoft Support: Apply data validation to cells; More on data validation; Create a drop-down list.
- Microsoft Support: Customize a slide master; What is a slide master; Apply/edit slide layouts.
- Microsoft Support: Add, change, or remove transitions; Animations and transitions; Animate text or objects.
- Microsoft Support: Save PowerPoint presentations as PDF.
- Microsoft Support: Guide to the Access user interface.

URLs (copy/paste):

<https://support.microsoft.com/en-us/office/word-features-comparison-web-vs-desktop-3e863ce3-e82c-4211-8f97-5b33c36c55f8>

<https://support.microsoft.com/en-us/office/basic-tasks-in-word-87b3243c-b0bf-4a29-82aa-09a681999fdc>

<https://support.microsoft.com/en-us/office/compare-and-merge-two-versions-of-a-document-f5059749-a797-4db7-a8fb-b3b27eb8b87e>

<https://support.microsoft.com/en-us/office/overview-of-pivottables-and-pivotcharts-527c8fa3-02c0-445a-a2db-7794676bce96>

<https://support.microsoft.com/en-us/office/create-a-pivottable-to-analyze-worksheet-data-a9a84538-bfe9-40a9-a8e9-f99134456576>

<https://support.microsoft.com/en-us/office/use-conditional-formatting-to-highlight-information-in-excel-fed60dfa-1d3f-4e13-9ecb-f1951ff89d7f>

<https://support.microsoft.com/en-us/office/apply-data-validation-to-cells-29fecbcc-d1b9-42c1-9d76-ef3ce5f6e1f0c4c>

<https://support.microsoft.com/en-us/office/more-on-data-validation-f38dee73-9900-4ca6-9301-8a5f6e1f0c4c>

<https://support.microsoft.com/en-us/office/create-a-drop-down-list-7693307a-59ef-400a-b769-c5402dce407b>

<https://support.microsoft.com/en-us/office/customize-a-slide-master-036d317b-3251-4237-8ddc-22f4668e2b56>

<https://support.microsoft.com/en-us/office/what-is-a-slide-master-in-powerpoint-b9abb2a0-7aef-4257-a14e-4329c904da54>

<https://support.microsoft.com/en-us/office/edit-and-re-apply-a-slide-layout-6f4338f8-555f-49cf-9835-6209be3c7b48>

<https://support.microsoft.com/en-us/office/add-change-or-remove-transitions-between-slides-3f8244bf-f893-4efd-a7eb-3a4845c9c971>

<https://support.microsoft.com/en-us/office/video-animations-and-transitions-for-powerpoint-web-07a941fb-b027-4f3e-b514-cd9866723f45>

<https://support.microsoft.com/en-us/office/animate-text-or-objects-305a1c94-83b1-4778-8df5-fcf7a9b7b7c6>

<https://support.microsoft.com/en-us/office/save-powerpoint-presentations-as-pdf-files-9b5c786b-9c6e-4fe6-81f6-9372f77c47c8>

<https://support.microsoft.com/en-us/office/guide-to-the-access-user-interface-cd8eca71-78a1-484f-897b-fc80b1ac77ec>

Appendix E. Screen-by-Screen Wireframes

Wireframes are intentionally low-fidelity. The implementation should follow these interaction patterns, not the exact pixels.

E.1 Document Library (Desktop)

```
+-----+
--+
| JS Office Suite                               Search...           [Import] [New]
|
+-----+
+-----+
| Filters          | Documents
|
| - All            | [Card] Writer: Project Plan      Updated 2m ago
|
| - Writer         | [Card] Sheets: Budget 2026          Updated yesterday
|
| - Sheets         | [Card] Slides: Deck                  Updated last week
|
| - Slides         | [Card] Base: CRM                    Updated 3d ago
|
| - Base          |
|
+-----+
+-----+
| Storage: 23% used | Tips: Ctrl/Cmd+K commands | Offline ✓ Autosave ✓
|
+-----+
--+
```

Notes:

- Keep actions close to the selection.
- Avoid modal spam.
- Always provide Undo.

E.2 New Document (Template Picker)

```
+-----+
| Create New                                     |
| App: (Writer) (Sheets) (Slides) (Base)       |
| Templates:                                   |
| [Blank] [Report] [Resume] [Newsletter] ...   |
| Name: _____                           |
| [Create]                                     [Cancel] |
+-----+
```

Notes:

- Keep actions close to the selection.
- Avoid modal spam.
- Always provide Undo.

E.3 Shared Settings

```
+-----+
| Settings                                     |
| Appearance: (Light/Dark/System) High Contrast [ ] |
| Motion: Reduced Motion [ ]                 |
| Storage: Used ____MB / Limit ____MB        |
| [Export Workspace] [Import Workspace] [Clear All Data] |
| Shortcuts: View / Customize (post-MVP)      |
+-----+
```

Notes:

- Keep actions close to the selection.
- Avoid modal spam.
- Always provide Undo.

E.4 Writer - Desktop

```
+-----+
--+
| Home | Insert | Layout | Review | View                               Doc Name
|
+-----+
--+
| [B] [I] [U]  Font ▼ Size ▼ Color ▼ Align ▼ Styles ▼  Comments ▼  Find ▼
|
+-----+
+-----+
| Nav pane (optional) | Page 1
|
| - Headings          | Lorem ipsum...
|
| - Comments          | [table] [image]
|
+-----+
+-----+
| Status: Page 1 of N | Words: 1,234 | Zoom 100% | Saved ✓
|
+-----+
--+
```

Notes:

- Keep actions close to the selection.
- Avoid modal spam.
- Always provide Undo.

E.5 Writer - Mobile

```
+-----+
| Doc Name ▼      Undo  Redo   Comments  ...  |
+-----+
| Chips: [B] [I] [U] [Styles] [Insert] [...] |
+-----+
| Paged editor surface                    |
| (tap to place cursor; drag handles)    |
+-----+
| Bottom: Library | Writer | Sheets | Slides |
+-----+
```

Notes:

- Keep actions close to the selection.
- Avoid modal spam.
- Always provide Undo.

E.6 Sheets - Desktop

```
+-----+
--+
| Name Box [A1]    fx [=SUM(A1:A10)]           [Calc: Auto] [Share*]
|
+-----+
--+
|      | A      B      C      D      E
|
|  1   |  12    30    44    ...
|
|  2   |  =A1*2  ...
|
|      | (virtualized grid)
|
+-----+
--+
| Tabs: [Sheet1] [+]    Status: Ready    Zoom 100%
|
+-----+
--+
```

Notes:

- Keep actions close to the selection.
- Avoid modal spam.
- Always provide Undo.

E.7 Sheets - Mobile

```
+-----+
| Sheet1 ▼   Cell A1   fx   Undo Redo   ...   |
+-----+
| Formula bar (expandable): =SUM(...)         |
+-----+
| Grid (pinch to zoom, drag handles)          |
+-----+
| Chips: Format | Data | Insert | Chart | ...   |
+-----+
```

Notes:

- Keep actions close to the selection.
- Avoid modal spam.
- Always provide Undo.

E.8 Slides - Desktop

```
+-----+
--+
| Home | Insert | Design | Transitions | Animations | Slide Show | View
|
+-----+
+-----+
| Thumbnails          | Canvas
|
| [1] [2] [3]         | [textbox] [shape] [image]
|
| [+]                 |
|
+-----+
+-----+
| Notes (collapsible) | Inspector (position, style)
|
+-----+
--+
```

Notes:

- Keep actions close to the selection.
- Avoid modal spam.
- Always provide Undo.

E.9 Slide Show / Presenter

+-----+	
Presenter (MVP - single screen)	
Current slide (large)	
Notes (below)	
Controls: Prev / Next / Timer (post-MVP)	
+-----+	

Notes:

- Keep actions close to the selection.
- Avoid modal spam.
- Always provide Undo.

E.10 Base - Navigation

+-----		
--+		
Base: CRM Demo	Search...	[New ▼]
+-----		
+-----+		
Tables	Customers (Datasheet)	
- Customers	[virtualized rows]	
- Orders		
Queries		
Forms		
Reports		
+-----		
+-----+		

Notes:

- Keep actions close to the selection.
- Avoid modal spam.
- Always provide Undo.

E.11 Base - Table Designer

```
+-----+
| Table Designer: Customers |
| Fields:                   |
|   ID (autonumber) [PK]   |
|   Name (text) [Required] |
|   Email (text) [Unique]  |
| Inspector: field properties (required, default, validation) |
+-----+
```

Notes:

- Keep actions close to the selection.
- Avoid modal spam.
- Always provide Undo.

E.12 Import/Export Wizard

```
+-----+
| Import                                     |
| Drop file here or [Choose file]          |
| Detected: Sheets workbook                |
| Options: (Create new) (Overwrite existing)|
| Preview: (read-only)                    |
| [Import]                                [Cancel] |
+-----+
```

Notes:

- Keep actions close to the selection.
- Avoid modal spam.
- Always provide Undo.

10. Master Task Checklist (authoritative)

Everything below is a checkbox task. Replace ☒ with ☐ when done and add a short verification note in parentheses e.g., '(verified: unit tests + manual run)'.

Shared Tasks

CORE

- ☐ CORE-001 [MVP] Bootstrap SPA shell with routing between Library/Writer/Sheets/Slides/Base.
- ☐ CORE-002 [MVP] Implement command registry (ID, label, icon, shortcut, enabled predicate, handler).
- ☐ CORE-003 [MVP] Implement command palette (Ctrl/Cmd+K) with fuzzy search and category filters.
- ☐ CORE-004 [MVP] Implement undo/redo framework (per-document history stacks) with batching.
- ☐ CORE-005 [MVP] Implement global selection model + focus management (keyboard safe).
- ☐ CORE-006 [MVP] Implement notifications/toasts + status bar messages.
- ☐ CORE-007 [MVP] Implement modal dialog system (confirm, prompt, file picker, error).
- ☐ CORE-008 [MVP] Implement theme system (light/dark + high-contrast) and design tokens.
- ☐ CORE-009 [MVP] Implement responsive layout system (breakpoints, safe areas, bottom sheets).
- ☐ CORE-010 [Post-MVP] Implement i18n scaffolding (string table, locale switch) (optional in MVP).

LIB

- ☐ LIB-001 [MVP] Document library screen: list/grid view with search and sort (recent, name, type).
- ☐ LIB-002 [MVP] Create document flow: choose app + template, name, create.
- ☐ LIB-003 [MVP] Recent documents section and pin/unpin.
- ☐ LIB-004 [MVP] Template gallery: built-in templates per app.
- ☐ LIB-005 [MVP] Document rename, duplicate, delete (with undo/restore).
- ☐ LIB-006 [MVP] Bulk export: export entire workspace as a JSON bundle.
- ☐ LIB-007 [MVP] Bulk import: import a workspace bundle; resolve conflicts (new copy vs overwrite).
- ☐ LIB-008 [MVP] Crash recovery screen: detect unsaved sessions and offer restore.
- ☐ LIB-009 [MVP] Settings: storage usage, clear cache, export all data, accessibility toggles.

STORAGE

- ☐ STORAGE-001 [MVP] StorageAdapter interface: get/set/remove/list keys; versioned namespaces per app.
- ☐ STORAGE-002 [MVP] localStorage implementation with chunking for large payloads (split strings).
- ☐ STORAGE-003 [MVP] Schema migration system: per-doc version field + migrators.
- ☐ STORAGE-004 [MVP] Autosave scheduler with debounce and 'dirty' tracking per document.
- ☐ STORAGE-005 [MVP] Snapshot history: rolling N snapshots per document with timestamps.
- ☐ STORAGE-006 [MVP] Import validation: JSON schema validator + size limits + graceful errors.
- ☐ STORAGE-007 [MVP] Export pipeline: deterministic JSON ordering + optional compression (base64).
- ☐ STORAGE-008 [MVP] Corruption detection: checksum per chunk + recovery to previous snapshot.
- ☐ STORAGE-009 [MVP] Cross-tab safety: detect concurrent edits in multiple tabs; warn and offer 'fork'.

EXPORT

- ☐ EXPORT-001 [MVP] JSON file export with download and copy-to-clipboard option.
- ☐ EXPORT-002 [MVP] JSON import via file picker and drag-drop onto library.
- ☐ EXPORT-003 [MVP] Per-doc 'Export summary' screen with metadata + preview thumbnail.
- ☐ EXPORT-004 [MVP] Per-doc 'Import preview' screen showing what will be created/overwritten.

A11Y

- ☐ A11Y-001 [MVP] Keyboard shortcuts framework (per-app + global) with conflict detection.

- ☐ A11Y-002 [MVP] ARIA roles for menus/toolbars/dialogs; focus trap in modals.
- ☐ A11Y-003 [MVP] Screen-reader announcements for selection changes and command results.
- ☐ A11Y-004 [MVP] Reduced motion mode: disables animations/transitions in UI and Slides playback.
- ☐ A11Y-005 [MVP] Touch selection handles component shared across apps.
- ☐ A11Y-006 [MVP] Pointer + keyboard friendly context menu component.

POLISH

- ☐ POLISH-001 [Post-MVP] Global search across document contents (post-MVP for large workspaces).
- ☐ POLISH-002 [MVP] Per-document version history viewer (snapshot list) and restore action.
- ☐ POLISH-003 [MVP] Thumbnail generation for library cards (Writer page 1, Sheets first sheet, Slides first slide, Base table list).
- ☐ POLISH-004 [MVP] Command hints: show shortcut labels in menus and tooltips.
- ☐ POLISH-005 [MVP] In-app help: searchable command reference and keyboard shortcut list.
- ☐ POLISH-006 [Post-MVP] Telemetry (post-MVP, privacy-respecting) for performance and crash reporting.

SHORTCUTS

- ☐ SHORTCUTS-001 [MVP] WRITER: Define default shortcut map for common commands (copy/paste/undo/redo/find).
- ☐ SHORTCUTS-002 [MVP] WRITER: Define app-specific shortcuts and ensure they don't conflict with browser defaults.
- ☐ SHORTCUTS-003 [Post-MVP] WRITER: Implement shortcut customization UI .
- ☐ SHORTCUTS-004 [MVP] SHEETS: Define default shortcut map for common commands (copy/paste/undo/redo/find).
- ☐ SHORTCUTS-005 [MVP] SHEETS: Define app-specific shortcuts and ensure they don't conflict with browser defaults.
- ☐ SHORTCUTS-006 [Post-MVP] SHEETS: Implement shortcut customization UI .
- ☐ SHORTCUTS-007 [MVP] SLIDES: Define default shortcut map for common commands (copy/paste/undo/redo/find).
- ☐ SHORTCUTS-008 [MVP] SLIDES: Define app-specific shortcuts and ensure they don't conflict with browser defaults.
- ☐ SHORTCUTS-009 [Post-MVP] SLIDES: Implement shortcut customization UI .
- ☐ SHORTCUTS-010 [MVP] BASE: Define default shortcut map for common commands (copy/paste/undo/redo/find).
- ☐ SHORTCUTS-011 [MVP] BASE: Define app-specific shortcuts and ensure they don't conflict with browser defaults.
- ☐ SHORTCUTS-012 [Post-MVP] BASE: Implement shortcut customization UI .

DOCS

- ☐ DOCS-001 [MVP] Write developer docs for CORE: data model, key components, and known pitfalls.
- ☐ DOCS-002 [MVP] Write user help for CORE: key workflows and tips.
- ☐ DOCS-003 [MVP] Write developer docs for LIB: data model, key components, and known pitfalls.
- ☐ DOCS-004 [MVP] Write user help for LIB: key workflows and tips.
- ☐ DOCS-005 [MVP] Write developer docs for STORAGE: data model, key components, and known pitfalls.
- ☐ DOCS-006 [MVP] Write user help for STORAGE: key workflows and tips.
- ☐ DOCS-007 [MVP] Write developer docs for A11Y: data model, key components, and known pitfalls.
- ☐ DOCS-008 [MVP] Write user help for A11Y: key workflows and tips.
- ☐ DOCS-009 [MVP] Write developer docs for EXPORT: data model, key components, and known pitfalls.
- ☐ DOCS-010 [MVP] Write user help for EXPORT: key workflows and tips.
- ☐ DOCS-011 [MVP] Write developer docs for WRITER-EDIT: data model, key components, and known pitfalls.
- ☐ DOCS-012 [MVP] Write user help for WRITER-EDIT: key workflows and tips.
- ☐ DOCS-013 [MVP] Write developer docs for WRITER-OBJECTS: data model, key components, and known pitfalls.

- ☐ DOCS-014 [MVP] Write user help for WRITER-OBJECTS: key workflows and tips.
- ☐ DOCS-015 [MVP] Write developer docs for WRITER-REVIEW: data model, key components, and known pitfalls.
- ☐ DOCS-016 [MVP] Write user help for WRITER-REVIEW: key workflows and tips.
- ☐ DOCS-017 [MVP] Write developer docs for WRITER-PRINT: data model, key components, and known pitfalls.
- ☐ DOCS-018 [MVP] Write user help for WRITER-PRINT: key workflows and tips.
- ☐ DOCS-019 [MVP] Write developer docs for SHEETS-GRID: data model, key components, and known pitfalls.
- ☐ DOCS-020 [MVP] Write user help for SHEETS-GRID: key workflows and tips.
- ☐ DOCS-021 [MVP] Write developer docs for SHEETS-CALC: data model, key components, and known pitfalls.
- ☐ DOCS-022 [MVP] Write user help for SHEETS-CALC: key workflows and tips.
- ☐ DOCS-023 [MVP] Write developer docs for SHEETS-FORMAT: data model, key components, and known pitfalls.
- ☐ DOCS-024 [MVP] Write user help for SHEETS-FORMAT: key workflows and tips.
- ☐ DOCS-025 [MVP] Write developer docs for SHEETS-DATA: data model, key components, and known pitfalls.
- ☐ DOCS-026 [MVP] Write user help for SHEETS-DATA: key workflows and tips.
- ☐ DOCS-027 [MVP] Write developer docs for SHEETS-CHART: data model, key components, and known pitfalls.
- ☐ DOCS-028 [MVP] Write user help for SHEETS-CHART: key workflows and tips.
- ☐ DOCS-029 [MVP] Write developer docs for SLIDES-CORE: data model, key components, and known pitfalls.
- ☐ DOCS-030 [MVP] Write user help for SLIDES-CORE: key workflows and tips.
- ☐ DOCS-031 [MVP] Write developer docs for SLIDES-TEXT: data model, key components, and known pitfalls.
- ☐ DOCS-032 [MVP] Write user help for SLIDES-TEXT: key workflows and tips.
- ☐ DOCS-033 [MVP] Write developer docs for SLIDES-SHAPES: data model, key components, and known pitfalls.
- ☐ DOCS-034 [MVP] Write user help for SLIDES-SHAPES: key workflows and tips.
- ☐ DOCS-035 [MVP] Write developer docs for SLIDES-MEDIA: data model, key components, and known pitfalls.
- ☐ DOCS-036 [MVP] Write user help for SLIDES-MEDIA: key workflows and tips.
- ☐ DOCS-037 [MVP] Write developer docs for SLIDES-MOTION: data model, key components, and known pitfalls.
- ☐ DOCS-038 [MVP] Write user help for SLIDES-MOTION: key workflows and tips.
- ☐ DOCS-039 [MVP] Write developer docs for SLIDES-SHOW: data model, key components, and known pitfalls.
- ☐ DOCS-040 [MVP] Write user help for SLIDES-SHOW: key workflows and tips.
- ☐ DOCS-041 [MVP] Write developer docs for BASE-NAV: data model, key components, and known pitfalls.
- ☐ DOCS-042 [MVP] Write user help for BASE-NAV: key workflows and tips.
- ☐ DOCS-043 [MVP] Write developer docs for BASE-TABLES: data model, key components, and known pitfalls.
- ☐ DOCS-044 [MVP] Write user help for BASE-TABLES: key workflows and tips.
- ☐ DOCS-045 [MVP] Write developer docs for BASE-REL: data model, key components, and known pitfalls.
- ☐ DOCS-046 [MVP] Write user help for BASE-REL: key workflows and tips.
- ☐ DOCS-047 [MVP] Write developer docs for BASE-QUERY: data model, key components, and known pitfalls.
- ☐ DOCS-048 [MVP] Write user help for BASE-QUERY: key workflows and tips.
- ☐ DOCS-049 [MVP] Write developer docs for BASE-FORMS: data model, key components, and known pitfalls.
- ☐ DOCS-050 [MVP] Write user help for BASE-FORMS: key workflows and tips.
- ☐ DOCS-051 [MVP] Write developer docs for BASE-REPORTS: data model, key components, and known pitfalls.
- ☐ DOCS-052 [MVP] Write user help for BASE-REPORTS: key workflows and tips.

QA

- ☐ QA-001 [MVP] Manual smoke test: Shell on Chrome (desktop) (create/edit/save/export/import).
- ☐ QA-002 [MVP] Accessibility smoke test: Shell on Chrome (desktop) (keyboard + screen reader basics).
- ☐ QA-003 [MVP] Manual smoke test: Shell on Firefox (desktop) (create/edit/save/export/import).
- ☐ QA-004 [MVP] Accessibility smoke test: Shell on Firefox (desktop) (keyboard + screen reader basics).
- ☐ QA-005 [MVP] Manual smoke test: Shell on Safari (desktop) (create/edit/save/export/import).
- ☐ QA-006 [MVP] Accessibility smoke test: Shell on Safari (desktop) (keyboard + screen reader basics).
- ☐ QA-007 [MVP] Manual smoke test: Shell on Chrome (Android) (create/edit/save/export/import).

- ☐ QA-008 [MVP] Accessibility smoke test: Shell on Chrome (Android) (keyboard + screen reader basics).
- ☐ QA-009 [MVP] Manual smoke test: Shell on Safari (iOS) (create/edit/save/export/import).
- ☐ QA-010 [MVP] Accessibility smoke test: Shell on Safari (iOS) (keyboard + screen reader basics).
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- ☐ QA-038 [MVP] Accessibility smoke test: Slides on Chrome (Android) (keyboard + screen reader basics).
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- ☐ QA-044 [MVP] Accessibility smoke test: Base on Firefox (desktop) (keyboard + screen reader basics).
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- ☐ QA-047 [MVP] Manual smoke test: Base on Chrome (Android) (create/edit/save/export/import).
- ☐ QA-048 [MVP] Accessibility smoke test: Base on Chrome (Android) (keyboard + screen reader basics).
- ☐ QA-049 [MVP] Manual smoke test: Base on Safari (iOS) (create/edit/save/export/import).
- ☐ QA-050 [MVP] Accessibility smoke test: Base on Safari (iOS) (keyboard + screen reader basics).

RELEASE

- ☐ RELEASE-001 [MVP] Write release checklist for M0 Shell (tests, perf, accessibility, backup).
- ☐ RELEASE-002 [MVP] Create demo сценарий for M0 Shell (scripted walkthrough).
- ☐ RELEASE-003 [MVP] Tag and archive build artifacts for M0 Shell.
- ☐ RELEASE-004 [MVP] Write release checklist for M1 Library+Storage (tests, perf, accessibility, backup).
- ☐ RELEASE-005 [MVP] Create demo сценарий for M1 Library+Storage (scripted walkthrough).
- ☐ RELEASE-006 [MVP] Tag and archive build artifacts for M1 Library+Storage.
- ☐ RELEASE-007 [MVP] Write release checklist for M2 Writer Slice (tests, perf, accessibility, backup).

- ☐ RELEASE-008 [MVP] Create demo сценарий for M2 Writer Slice (scripted walkthrough).
- ☐ RELEASE-009 [MVP] Tag and archive build artifacts for M2 Writer Slice.
- ☐ RELEASE-010 [MVP] Write release checklist for M3 Sheets Slice (tests, perf, accessibility, backup).
- ☐ RELEASE-011 [MVP] Create demo сценарий for M3 Sheets Slice (scripted walkthrough).
- ☐ RELEASE-012 [MVP] Tag and archive build artifacts for M3 Sheets Slice.
- ☐ RELEASE-013 [MVP] Write release checklist for M4 Slides Slice (tests, perf, accessibility, backup).
- ☐ RELEASE-014 [MVP] Create demo сценарий for M4 Slides Slice (scripted walkthrough).
- ☐ RELEASE-015 [MVP] Tag and archive build artifacts for M4 Slides Slice.
- ☐ RELEASE-016 [MVP] Write release checklist for M5 Base Slice (tests, perf, accessibility, backup).
- ☐ RELEASE-017 [MVP] Create demo сценарий for M5 Base Slice (scripted walkthrough).
- ☐ RELEASE-018 [MVP] Tag and archive build artifacts for M5 Base Slice.
- ☐ RELEASE-019 [MVP] Write release checklist for M6 Hardening (tests, perf, accessibility, backup).
- ☐ RELEASE-020 [MVP] Create demo сценарий for M6 Hardening (scripted walkthrough).
- ☐ RELEASE-021 [MVP] Tag and archive build artifacts for M6 Hardening.

CLOUD

- ☐ CLOUD-001 [Post-MVP] Design cloud save API and auth (post-MVP).
- ☐ CLOUD-002 [Post-MVP] Implement sync engine with conflict resolution (post-MVP).
- ☐ CLOUD-003 [Post-MVP] Optional collaboration via CRDT (post-MVP).

Writer Tasks

WRITER-EDIT

- ☐ WRITER-EDIT-001 [MVP] Editor core: structured model editing (blocks + inline runs) with cursor mapping.
- ☐ WRITER-EDIT-002 [MVP] Basic text input, deletion, selection, multi-paragraph selection.
- ☐ WRITER-EDIT-003 [MVP] Inline formatting commands: bold/italic/underline/strike.
- ☐ WRITER-EDIT-004 [MVP] Font family, size, color, highlight; clear formatting.
- ☐ WRITER-EDIT-005 [MVP] Paragraph formatting: align left/center/right/justify.
- ☐ WRITER-EDIT-006 [MVP] Paragraph spacing + line spacing controls (basic).
- ☐ WRITER-EDIT-007 [MVP] Indent/outdent and tab stops (basic).
- ☐ WRITER-EDIT-008 [MVP] Bullets and numbering; toggle list; continue list on Enter.
- ☐ WRITER-EDIT-009 [MVP] Multilevel lists (basic): Tab/Shift-Tab to indent/outdent list level.
- ☐ WRITER-EDIT-010 [MVP] Styles: apply built-in styles; show current style; style gallery.
- ☐ WRITER-EDIT-011 [MVP] Find/replace: plain text, case sensitive toggle, next/prev navigation.
- ☐ WRITER-EDIT-012 [MVP] Word/character count and estimated reading time (optional).
- ☐ WRITER-EDIT-013 [Post-MVP] Rulers (optional): show/hide; adjust margins/indents.

WRITER-OBJECTS

- ☐ WRITER-OBJECTS-001 [MVP] Insert image: file picker, paste, drag-drop; store as data URL or blob reference.
- ☐ WRITER-OBJECTS-002 [MVP] Image resize handles; lock aspect ratio; set alt text.
- ☐ WRITER-OBJECTS-003 [Post-MVP] Text wrap modes: inline + square (basic); advanced wrapping.
- ☐ WRITER-OBJECTS-004 [MVP] Insert hyperlink: create/edit/remove; Ctrl/Cmd+Click to open.
- ☐ WRITER-OBJECTS-005 [MVP] Insert table: choose rows/cols; render with selection.
- ☐ WRITER-OBJECTS-006 [MVP] Table editing: add/remove rows/cols, merge/split cells (basic).
- ☐ WRITER-OBJECTS-007 [MVP] Table formatting: borders, shading, cell padding (basic).
- ☐ WRITER-OBJECTS-008 [MVP] Headers/footers editor with distinct area; page number placeholder.
- ☐ WRITER-OBJECTS-009 [MVP] Page breaks: insert/remove; show page boundaries.
- ☐ WRITER-OBJECTS-010 [Post-MVP] Columns: section-based columns with rules.

WRITER-REVIEW

- ☐ WRITER-REVIEW-001 [MVP] Comments: add comment anchored to text range; threaded replies; resolve/reopen.
- ☐ WRITER-REVIEW-002 [MVP] Comment sidebar with navigation to anchors.
- ☐ WRITER-REVIEW-003 [MVP] Spellcheck: integrate browser spellcheck; custom dictionary list per workspace.
- ☐ WRITER-REVIEW-004 [Post-MVP] Track changes: record insert/delete/format changes; accept/reject flows.
- ☐ WRITER-REVIEW-005 [Post-MVP] Compare/merge: import second version; compute diff and show revision marks.

WRITER-PRINT

- ☐ WRITER-PRINT-001 [MVP] Print/PDF: print stylesheet; page headers/footers; basic pagination.
- ☐ WRITER-PRINT-002 [Post-MVP] Export to PDF: deterministic PDF generation engine (optional).
- ☐ WRITER-PRINT-003 [MVP] Zoom controls: fit page, fit width, 100%, custom.
- ☐ WRITER-PRINT-004 [Post-MVP] View modes: print layout (default), web layout, outline.

WRITER-FEAT

- ☐ WRITER-FEAT-001 [Post-MVP] Superscript/subscript formatting.
- ☐ WRITER-FEAT-002 [Post-MVP] Text case tools: UPPER/lower/Title Case.
- ☐ WRITER-FEAT-003 [Post-MVP] Paragraph borders and shading.
- ☐ WRITER-FEAT-004 [Post-MVP] Section breaks and per-section headers/footers.
- ☐ WRITER-FEAT-005 [Post-MVP] Footnotes and endnotes.

- ☐ WRITER-FEAT-006 [Post-MVP] Automatic Table of Contents generation from headings.
- ☐ WRITER-FEAT-007 [Post-MVP] Insert symbols and special characters palette.
- ☐ WRITER-FEAT-008 [Post-MVP] Equation editor (basic LaTeX input) (optional).
- ☐ WRITER-FEAT-009 [Post-MVP] Columns layout (newspaper columns).
- ☐ WRITER-FEAT-010 [Post-MVP] Page numbers in multiple formats (roman, start at N).
- ☐ WRITER-FEAT-011 [Post-MVP] Watermarks and page background color.
- ☐ WRITER-FEAT-012 [Post-MVP] Text box objects (floating) (post-MVP).
- ☐ WRITER-FEAT-013 [Post-MVP] Shapes in Writer (post-MVP).
- ☐ WRITER-FEAT-014 [Post-MVP] Document protection/restrict editing (post-MVP).
- ☐ WRITER-FEAT-015 [Post-MVP] References: citations + bibliography manager (post-MVP).

WRITER-CMDS

- ☐ WRITER-CMDS-001 [MVP] Writer command: Increase/decrease font size.
- ☐ WRITER-CMDS-002 [MVP] Writer command: Change font color.
- ☐ WRITER-CMDS-003 [MVP] Writer command: Text highlight color.
- ☐ WRITER-CMDS-004 [MVP] Writer command: Clear formatting.
- ☐ WRITER-CMDS-005 [MVP] Writer command: Increase/decrease indent.
- ☐ WRITER-CMDS-006 [MVP] Writer command: Line spacing presets.
- ☐ WRITER-CMDS-007 [Post-MVP] Writer command: Paragraph spacing before/after.
- ☐ WRITER-CMDS-008 [Post-MVP] Writer command: Border presets.
- ☐ WRITER-CMDS-009 [Post-MVP] Writer command: Shading presets.
- ☐ WRITER-CMDS-010 [Post-MVP] Writer command: Page orientation portrait/landscape.
- ☐ WRITER-CMDS-011 [Post-MVP] Writer command: Margins presets (Normal/Narrow/Wide).
- ☐ WRITER-CMDS-012 [Post-MVP] Writer command: Columns (2/3).
- ☐ WRITER-CMDS-013 [Post-MVP] Writer command: Insert page number.
- ☐ WRITER-CMDS-014 [Post-MVP] Writer command: Insert date/time.
- ☐ WRITER-CMDS-015 [Post-MVP] Writer command: Insert horizontal line.
- ☐ WRITER-CMDS-016 [Post-MVP] Writer command: Insert table of contents.
- ☐ WRITER-CMDS-017 [Post-MVP] Writer command: Insert footnote.
- ☐ WRITER-CMDS-018 [Post-MVP] Writer command: Insert endnote.

WRITER-RIBBON

- ☐ WRITER-RIBBON-001 [Post-MVP] Writer ribbon: Clipboard -> Paste special.
- ☐ WRITER-RIBBON-002 [Post-MVP] Writer ribbon: Clipboard -> Format painter.
- ☐ WRITER-RIBBON-003 [MVP] Writer ribbon: Font -> Bold/Italic/Underline.
- ☐ WRITER-RIBBON-004 [MVP] Writer ribbon: Font -> Strikethrough.
- ☐ WRITER-RIBBON-005 [Post-MVP] Writer ribbon: Font -> Subscript.
- ☐ WRITER-RIBBON-006 [Post-MVP] Writer ribbon: Font -> Superscript.
- ☐ WRITER-RIBBON-007 [Post-MVP] Writer ribbon: Font -> Text effects.
- ☐ WRITER-RIBBON-008 [Post-MVP] Writer ribbon: Font -> Change case.
- ☐ WRITER-RIBBON-009 [MVP] Writer ribbon: Font -> Clear formatting.
- ☐ WRITER-RIBBON-010 [MVP] Writer ribbon: Paragraph -> Bullets.
- ☐ WRITER-RIBBON-011 [MVP] Writer ribbon: Paragraph -> Numbering.
- ☐ WRITER-RIBBON-012 [MVP] Writer ribbon: Paragraph -> Multilevel list.
- ☐ WRITER-RIBBON-013 [MVP] Writer ribbon: Paragraph -> Decrease/Increase indent.
- ☐ WRITER-RIBBON-014 [Post-MVP] Writer ribbon: Paragraph -> Sort paragraphs.
- ☐ WRITER-RIBBON-015 [Post-MVP] Writer ribbon: Paragraph -> Show/hide formatting marks.
- ☐ WRITER-RIBBON-016 [MVP] Writer ribbon: Paragraph -> Line and paragraph spacing.
- ☐ WRITER-RIBBON-017 [Post-MVP] Writer ribbon: Paragraph -> Borders/Shading.
- ☐ WRITER-RIBBON-018 [MVP] Writer ribbon: Styles -> Styles gallery.
- ☐ WRITER-RIBBON-019 [Post-MVP] Writer ribbon: Styles -> Create/modify style.
- ☐ WRITER-RIBBON-020 [Post-MVP] Writer ribbon: Styles -> Style inspector.

- ☐ WRITER-RIBBON-021 [Post-MVP] Writer ribbon: Insert -> Cover page.
- ☐ WRITER-RIBBON-022 [Post-MVP] Writer ribbon: Insert -> Blank page.
- ☐ WRITER-RIBBON-023 [MVP] Writer ribbon: Insert -> Page break.
- ☐ WRITER-RIBBON-024 [MVP] Writer ribbon: Insert -> Table.
- ☐ WRITER-RIBBON-025 [MVP] Writer ribbon: Insert -> Pictures.
- ☐ WRITER-RIBBON-026 [Post-MVP] Writer ribbon: Insert -> Shapes.
- ☐ WRITER-RIBBON-027 [Post-MVP] Writer ribbon: Insert -> Icons.
- ☐ WRITER-RIBBON-028 [MVP] Writer ribbon: Insert -> Links.
- ☐ WRITER-RIBBON-029 [MVP] Writer ribbon: Insert -> Header.
- ☐ WRITER-RIBBON-030 [MVP] Writer ribbon: Insert -> Footer.
- ☐ WRITER-RIBBON-031 [Post-MVP] Writer ribbon: Insert -> Page number.
- ☐ WRITER-RIBBON-032 [Post-MVP] Writer ribbon: Insert -> Text box.
- ☐ WRITER-RIBBON-033 [Post-MVP] Writer ribbon: Insert -> WordArt.
- ☐ WRITER-RIBBON-034 [Post-MVP] Writer ribbon: Insert -> Drop cap.
- ☐ WRITER-RIBBON-035 [Post-MVP] Writer ribbon: Insert -> Signature line.
- ☐ WRITER-RIBBON-036 [Post-MVP] Writer ribbon: Insert -> Equation.
- ☐ WRITER-RIBBON-037 [Post-MVP] Writer ribbon: Insert -> Symbol.
- ☐ WRITER-RIBBON-038 [Post-MVP] Writer ribbon: Layout -> Margins.
- ☐ WRITER-RIBBON-039 [Post-MVP] Writer ribbon: Layout -> Orientation.
- ☐ WRITER-RIBBON-040 [Post-MVP] Writer ribbon: Layout -> Size.
- ☐ WRITER-RIBBON-041 [Post-MVP] Writer ribbon: Layout -> Columns.
- ☐ WRITER-RIBBON-042 [Post-MVP] Writer ribbon: Layout -> Breaks.
- ☐ WRITER-RIBBON-043 [Post-MVP] Writer ribbon: Layout -> Line numbers.
- ☐ WRITER-RIBBON-044 [Post-MVP] Writer ribbon: Layout -> Hyphenation.
- ☐ WRITER-RIBBON-045 [Post-MVP] Writer ribbon: References -> Table of Contents.
- ☐ WRITER-RIBBON-046 [Post-MVP] Writer ribbon: References -> Footnotes.
- ☐ WRITER-RIBBON-047 [Post-MVP] Writer ribbon: References -> Citations & Bibliography.
- ☐ WRITER-RIBBON-048 [Post-MVP] Writer ribbon: References -> Captions.
- ☐ WRITER-RIBBON-049 [Post-MVP] Writer ribbon: References -> Index.
- ☐ WRITER-RIBBON-050 [MVP] Writer ribbon: Review -> Spelling & Grammar.
- ☐ WRITER-RIBBON-051 [Post-MVP] Writer ribbon: Review -> Thesaurus.
- ☐ WRITER-RIBBON-052 [MVP] Writer ribbon: Review -> Word count.
- ☐ WRITER-RIBBON-053 [MVP] Writer ribbon: Review -> New Comment.
- ☐ WRITER-RIBBON-054 [Post-MVP] Writer ribbon: Review -> Track Changes.
- ☐ WRITER-RIBBON-055 [Post-MVP] Writer ribbon: Review -> Accept/Reject.
- ☐ WRITER-RIBBON-056 [Post-MVP] Writer ribbon: Review -> Compare.
- ☐ WRITER-RIBBON-057 [Post-MVP] Writer ribbon: Review -> Restrict Editing.
- ☐ WRITER-RIBBON-058 [MVP] Writer ribbon: View -> Zoom controls.
- ☐ WRITER-RIBBON-059 [Post-MVP] Writer ribbon: View -> Ruler.
- ☐ WRITER-RIBBON-060 [Post-MVP] Writer ribbon: View -> Navigation pane.
- ☐ WRITER-RIBBON-061 [Post-MVP] Writer ribbon: View -> Outline view.
- ☐ WRITER-RIBBON-062 [Post-MVP] Writer ribbon: View -> Immersive reader.

WRITER-TEST

- ☐ WRITER-TEST-001 [MVP] Integration tests for WRITER-EDIT: typing, selection, formatting, undo/redo, persistence.
- ☐ WRITER-TEST-002 [MVP] Accessibility smoke tests for WRITER-EDIT: keyboard-only flows and ARIA checks.
- ☐ WRITER-TEST-003 [MVP] Integration tests for WRITER-OBJECTS: typing, selection, formatting, undo/redo, persistence.
- ☐ WRITER-TEST-004 [MVP] Accessibility smoke tests for WRITER-OBJECTS: keyboard-only flows and ARIA checks.

- ☐ WRITER-TEST-005 [MVP] Integration tests for WRITER-REVIEW: typing, selection, formatting, undo/redo, persistence.
- ☐ WRITER-TEST-006 [MVP] Accessibility smoke tests for WRITER-REVIEW: keyboard-only flows and ARIA checks.
- ☐ WRITER-TEST-007 [MVP] Integration tests for WRITER-PRINT: typing, selection, formatting, undo/redo, persistence.
- ☐ WRITER-TEST-008 [MVP] Accessibility smoke tests for WRITER-PRINT: keyboard-only flows and ARIA checks.

FORMATS-DOCX

- ☐ FORMATS-DOCX-001 [Post-MVP] DOCX import: paragraphs + basic styles.
- ☐ FORMATS-DOCX-002 [Post-MVP] DOCX import: tables.
- ☐ FORMATS-DOCX-003 [Post-MVP] DOCX import: images.
- ☐ FORMATS-DOCX-004 [Post-MVP] DOCX export: paragraphs + basic styles.
- ☐ FORMATS-DOCX-005 [Post-MVP] DOCX export: tables.
- ☐ FORMATS-DOCX-006 [Post-MVP] DOCX export: images.
- ☐ FORMATS-DOCX-007 [Post-MVP] DOCX export: headers/footers.

Sheets Tasks

SHEETS-GRID

- ☐ SHEETS-GRID-001 [MVP] Virtualized grid renderer (rows/cols) with smooth scrolling.
- ☐ SHEETS-GRID-002 [MVP] Row/column headers: selection, resize, hide/show, insert/delete.
- ☐ SHEETS-GRID-003 [Post-MVP] Cell selection model: single cell, range, multi-range, keyboard navigation.
- ☐ SHEETS-GRID-004 [MVP] Cell editor: in-cell edit + formula bar edit; commit/cancel behaviors.
- ☐ SHEETS-GRID-005 [MVP] Clipboard: copy/paste ranges; preserve formulas and formats (basic).
- ☐ SHEETS-GRID-006 [MVP] Fill handle: drag to autofill series (basic patterns).
- ☐ SHEETS-GRID-007 [Post-MVP] Freeze panes: top/left arbitrary; split view.
- ☐ SHEETS-GRID-008 [Post-MVP] Sheet tabs: add/rename/duplicate/move; color tabs.
- ☐ SHEETS-GRID-009 [MVP] Find/replace across sheet; highlight matches.
- ☐ SHEETS-GRID-010 [MVP] Undo/redo for grid edits and formatting.

SHEETS-CALC

- ☐ SHEETS-CALC-001 [MVP] Formula lexer/parser to AST supporting operators +, -, *, /, ^ and parentheses.
- ☐ SHEETS-CALC-002 [MVP] Reference resolver: relative/absolute refs; cross-sheet refs; range refs.
- ☐ SHEETS-CALC-003 [MVP] Dependency graph builder and incremental recalculation.
- ☐ SHEETS-CALC-004 [MVP] Error propagation (#DIV/0!, #VALUE!, #REF!, #NAME?, #N/A).
- ☐ SHEETS-CALC-005 [MVP] Function library MVP: SUM, AVERAGE, MIN, MAX, COUNT, COUNTA, IF, AND, OR, NOT.
- ☐ SHEETS-CALC-006 [MVP] Function library MVP: CONCAT, LEFT, RIGHT, MID, LEN, TRIM, LOWER, UPPER.
- ☐ SHEETS-CALC-007 [MVP] Function library MVP: TODAY, NOW, DATE, YEAR, MONTH, DAY.
- ☐ SHEETS-CALC-008 [Post-MVP] Lookup MVP: VLOOKUP (exact match); XLOOKUP, INDEX, MATCH.
- ☐ SHEETS-CALC-009 [MVP] Volatile function scheduling (NOW/TODAY) and recalc triggers.

SHEETS-FORMAT

- ☐ SHEETS-FORMAT-001 [MVP] Cell formatting: font, size, bold/italic/underline, text color, fill.
- ☐ SHEETS-FORMAT-002 [MVP] Number formats: general, number, currency, percent, date, time (basic).
- ☐ SHEETS-FORMAT-003 [MVP] Alignment: left/center/right, vertical align (basic), wrap text.
- ☐ SHEETS-FORMAT-004 [MVP] Borders: all/inner/outer; border color (basic).
- ☐ SHEETS-FORMAT-005 [MVP] Conditional formatting MVP: simple rules (greater than, less than, between) with colors.
- ☐ SHEETS-FORMAT-006 [Post-MVP] Conditional formatting: icon sets, data bars, formula-based rules.
- ☐ SHEETS-FORMAT-007 [MVP] Data validation MVP: list dropdown + numeric range + date range.
- ☐ SHEETS-FORMAT-008 [MVP] Data validation: input message and error alert (basic).
- ☐ SHEETS-FORMAT-009 [MVP] Cell comments/notes (optional in MVP).

SHEETS-DATA

- ☐ SHEETS-DATA-001 [Post-MVP] Sort: single column ascending/descending; multi-column sort.
- ☐ SHEETS-DATA-002 [MVP] Filter: header dropdown filter; filter by value/search (basic).
- ☐ SHEETS-DATA-003 [MVP] Remove duplicates (basic) for selected range.
- ☐ SHEETS-DATA-004 [Post-MVP] Text-to-columns.
- ☐ SHEETS-DATA-005 [MVP] Import CSV wizard: delimiter, header row, column types.
- ☐ SHEETS-DATA-006 [MVP] Export CSV per sheet with proper escaping and newlines.
- ☐ SHEETS-DATA-007 [MVP] Named ranges MVP: create/edit/delete; use in formulas.
- ☐ SHEETS-DATA-008 [Post-MVP] Tables: structured tables, auto filters, structured refs.
- ☐ SHEETS-DATA-009 [Post-MVP] Pivot tables: field list, rows/cols/values/filters, grouping.

SHEETS-CHART

- ☐ SHEETS-CHART-001 [MVP] Chart object model and rendering (SVG preferred).
- ☐ SHEETS-CHART-002 [MVP] Insert chart from selected range: bar/line/pie/scatter.
- ☐ SHEETS-CHART-003 [MVP] Chart editor: change type, series names, legend, title.
- ☐ SHEETS-CHART-004 [Post-MVP] Chart formatting (basic): axis labels, gridlines (post-MVP advanced).
- ☐ SHEETS-CHART-005 [MVP] Embed charts in sheet and allow resizing/moving.

SHEETS-CHART-TYPES

- ☐ SHEETS-CHART-TYPES-001 [MVP] Support chart type: Column.
- ☐ SHEETS-CHART-TYPES-002 [MVP] Support chart type: Bar.
- ☐ SHEETS-CHART-TYPES-003 [MVP] Support chart type: Line.
- ☐ SHEETS-CHART-TYPES-004 [Post-MVP] Support chart type: Area.
- ☐ SHEETS-CHART-TYPES-005 [MVP] Support chart type: Pie.
- ☐ SHEETS-CHART-TYPES-006 [Post-MVP] Support chart type: Doughnut.
- ☐ SHEETS-CHART-TYPES-007 [MVP] Support chart type: Scatter.
- ☐ SHEETS-CHART-TYPES-008 [Post-MVP] Support chart type: Bubble.
- ☐ SHEETS-CHART-TYPES-009 [Post-MVP] Support chart type: Radar.
- ☐ SHEETS-CHART-TYPES-010 [Post-MVP] Support chart type: Combo.
- ☐ SHEETS-CHART-TYPES-011 [Post-MVP] Support chart type: Histogram.
- ☐ SHEETS-CHART-TYPES-012 [Post-MVP] Support chart type: Box & Whisker.
- ☐ SHEETS-CHART-TYPES-013 [Post-MVP] Support chart type: Waterfall.
- ☐ SHEETS-CHART-TYPES-014 [Post-MVP] Support chart type: Treemap.
- ☐ SHEETS-CHART-TYPES-015 [Post-MVP] Support chart type: Sunburst.

SHEETS-RIBBON

- ☐ SHEETS-RIBBON-001 [Post-MVP] Sheets ribbon: Home -> Paste special.
- ☐ SHEETS-RIBBON-002 [Post-MVP] Sheets ribbon: Home -> Format painter.
- ☐ SHEETS-RIBBON-003 [MVP] Sheets ribbon: Home -> Number formats.
- ☐ SHEETS-RIBBON-004 [MVP] Sheets ribbon: Home -> Conditional formatting.
- ☐ SHEETS-RIBBON-005 [Post-MVP] Sheets ribbon: Home -> Format as table.
- ☐ SHEETS-RIBBON-006 [Post-MVP] Sheets ribbon: Home -> AutoSum.
- ☐ SHEETS-RIBBON-007 [MVP] Sheets ribbon: Home -> Fill.
- ☐ SHEETS-RIBBON-008 [MVP] Sheets ribbon: Home -> Clear.
- ☐ SHEETS-RIBBON-009 [MVP] Sheets ribbon: Home -> Sort & Filter.
- ☐ SHEETS-RIBBON-010 [MVP] Sheets ribbon: Home -> Find & Select.
- ☐ SHEETS-RIBBON-011 [Post-MVP] Sheets ribbon: Insert -> PivotTable.
- ☐ SHEETS-RIBBON-012 [Post-MVP] Sheets ribbon: Insert -> Recommended charts.
- ☐ SHEETS-RIBBON-013 [MVP] Sheets ribbon: Insert -> Chart types gallery.
- ☐ SHEETS-RIBBON-014 [Post-MVP] Sheets ribbon: Insert -> Sparklines.
- ☐ SHEETS-RIBBON-015 [Post-MVP] Sheets ribbon: Insert -> Slicer.
- ☐ SHEETS-RIBBON-016 [Post-MVP] Sheets ribbon: Insert -> Timeline.
- ☐ SHEETS-RIBBON-017 [MVP] Sheets ribbon: Insert -> Link.
- ☐ SHEETS-RIBBON-018 [Post-MVP] Sheets ribbon: Formulas -> Insert function dialog.
- ☐ SHEETS-RIBBON-019 [Post-MVP] Sheets ribbon: Formulas -> Name manager.
- ☐ SHEETS-RIBBON-020 [Post-MVP] Sheets ribbon: Formulas -> Formula auditing (trace precedents/dependents).
- ☐ SHEETS-RIBBON-021 [Post-MVP] Sheets ribbon: Formulas -> Calculation options (auto/manual).
- ☐ SHEETS-RIBBON-022 [MVP] Sheets ribbon: Data -> Get data/import CSV.
- ☐ SHEETS-RIBBON-023 [Post-MVP] Sheets ribbon: Data -> Text to columns.
- ☐ SHEETS-RIBBON-024 [MVP] Sheets ribbon: Data -> Remove duplicates.
- ☐ SHEETS-RIBBON-025 [MVP] Sheets ribbon: Data -> Data validation.
- ☐ SHEETS-RIBBON-026 [Post-MVP] Sheets ribbon: Data -> What-if analysis.

- ☐ SHEETS-RIBBON-027 [Post-MVP] Sheets ribbon: Data -> Forecast sheet.
- ☐ SHEETS-RIBBON-028 [Post-MVP] Sheets ribbon: Review -> New comment/notes.
- ☐ SHEETS-RIBBON-029 [Post-MVP] Sheets ribbon: Review -> Protect sheet/workbook.
- ☐ SHEETS-RIBBON-030 [MVP] Sheets ribbon: View -> Freeze panes.
- ☐ SHEETS-RIBBON-031 [Post-MVP] Sheets ribbon: View -> Split view.
- ☐ SHEETS-RIBBON-032 [Post-MVP] Sheets ribbon: View -> Show gridlines/headings.
- ☐ SHEETS-RIBBON-033 [MVP] Sheets ribbon: View -> Zoom.

SHEETS-FUNC-MATH

- ☐ SHEETS-FUNC-MATH-001 [MVP] Implement spreadsheet function: SUM.
- ☐ SHEETS-FUNC-MATH-002 [Post-MVP] Implement spreadsheet function: SUMIF.
- ☐ SHEETS-FUNC-MATH-003 [Post-MVP] Implement spreadsheet function: SUMIFS.
- ☐ SHEETS-FUNC-MATH-004 [Post-MVP] Implement spreadsheet function: PRODUCT.
- ☐ SHEETS-FUNC-MATH-005 [Post-MVP] Implement spreadsheet function: SUBTOTAL.
- ☐ SHEETS-FUNC-MATH-006 [Post-MVP] Implement spreadsheet function: ROUND.
- ☐ SHEETS-FUNC-MATH-007 [Post-MVP] Implement spreadsheet function: ROUNDUP.
- ☐ SHEETS-FUNC-MATH-008 [Post-MVP] Implement spreadsheet function: ROUNDDOWN.
- ☐ SHEETS-FUNC-MATH-009 [Post-MVP] Implement spreadsheet function: INT.
- ☐ SHEETS-FUNC-MATH-010 [Post-MVP] Implement spreadsheet function: MOD.
- ☐ SHEETS-FUNC-MATH-011 [Post-MVP] Implement spreadsheet function: POWER.
- ☐ SHEETS-FUNC-MATH-012 [Post-MVP] Implement spreadsheet function: SQRT.
- ☐ SHEETS-FUNC-MATH-013 [Post-MVP] Implement spreadsheet function: ABS.
- ☐ SHEETS-FUNC-MATH-014 [Post-MVP] Implement spreadsheet function: CEILING.
- ☐ SHEETS-FUNC-MATH-015 [Post-MVP] Implement spreadsheet function: FLOOR.
- ☐ SHEETS-FUNC-MATH-016 [Post-MVP] Implement spreadsheet function: RAND.
- ☐ SHEETS-FUNC-MATH-017 [Post-MVP] Implement spreadsheet function: RANDBETWEEN.

SHEETS-FUNC-STAT

- ☐ SHEETS-FUNC-STAT-001 [MVP] Implement spreadsheet function: AVERAGE.
- ☐ SHEETS-FUNC-STAT-002 [Post-MVP] Implement spreadsheet function: AVERAGEIF.
- ☐ SHEETS-FUNC-STAT-003 [Post-MVP] Implement spreadsheet function: AVERAGEIFS.
- ☐ SHEETS-FUNC-STAT-004 [MVP] Implement spreadsheet function: COUNT.
- ☐ SHEETS-FUNC-STAT-005 [MVP] Implement spreadsheet function: COUNTA.
- ☐ SHEETS-FUNC-STAT-006 [Post-MVP] Implement spreadsheet function: COUNTIF.
- ☐ SHEETS-FUNC-STAT-007 [Post-MVP] Implement spreadsheet function: COUNTIFS.
- ☐ SHEETS-FUNC-STAT-008 [Post-MVP] Implement spreadsheet function: MEDIAN.
- ☐ SHEETS-FUNC-STAT-009 [Post-MVP] Implement spreadsheet function: MODE.
- ☐ SHEETS-FUNC-STAT-010 [Post-MVP] Implement spreadsheet function: STDEV.S.
- ☐ SHEETS-FUNC-STAT-011 [Post-MVP] Implement spreadsheet function: STDEV.P.
- ☐ SHEETS-FUNC-STAT-012 [Post-MVP] Implement spreadsheet function: VAR.S.
- ☐ SHEETS-FUNC-STAT-013 [Post-MVP] Implement spreadsheet function: VAR.P.
- ☐ SHEETS-FUNC-STAT-014 [Post-MVP] Implement spreadsheet function: PERCENTILE.
- ☐ SHEETS-FUNC-STAT-015 [Post-MVP] Implement spreadsheet function: QUARTILE.

SHEETS-FUNC-LOGIC

- ☐ SHEETS-FUNC-LOGIC-001 [MVP] Implement spreadsheet function: IF.
- ☐ SHEETS-FUNC-LOGIC-002 [Post-MVP] Implement spreadsheet function: IFS.
- ☐ SHEETS-FUNC-LOGIC-003 [Post-MVP] Implement spreadsheet function: SWITCH.
- ☐ SHEETS-FUNC-LOGIC-004 [MVP] Implement spreadsheet function: AND.
- ☐ SHEETS-FUNC-LOGIC-005 [MVP] Implement spreadsheet function: OR.
- ☐ SHEETS-FUNC-LOGIC-006 [MVP] Implement spreadsheet function: NOT.
- ☐ SHEETS-FUNC-LOGIC-007 [Post-MVP] Implement spreadsheet function: IFERROR.

- ☐ SHEETS-FUNC-LOGIC-008 [Post-MVP] Implement spreadsheet function: IFNA.
- ☐ SHEETS-FUNC-LOGIC-009 [Post-MVP] Implement spreadsheet function: XOR.

SHEETS-FUNC-TEXT

- ☐ SHEETS-FUNC-TEXT-001 [MVP] Implement spreadsheet function: CONCAT.
- ☐ SHEETS-FUNC-TEXT-002 [Post-MVP] Implement spreadsheet function: TEXTJOIN.
- ☐ SHEETS-FUNC-TEXT-003 [MVP] Implement spreadsheet function: LEFT.
- ☐ SHEETS-FUNC-TEXT-004 [MVP] Implement spreadsheet function: RIGHT.
- ☐ SHEETS-FUNC-TEXT-005 [MVP] Implement spreadsheet function: MID.
- ☐ SHEETS-FUNC-TEXT-006 [MVP] Implement spreadsheet function: LEN.
- ☐ SHEETS-FUNC-TEXT-007 [MVP] Implement spreadsheet function: TRIM.
- ☐ SHEETS-FUNC-TEXT-008 [Post-MVP] Implement spreadsheet function: CLEAN.
- ☐ SHEETS-FUNC-TEXT-009 [MVP] Implement spreadsheet function: LOWER.
- ☐ SHEETS-FUNC-TEXT-010 [MVP] Implement spreadsheet function: UPPER.
- ☐ SHEETS-FUNC-TEXT-011 [Post-MVP] Implement spreadsheet function: PROPER.
- ☐ SHEETS-FUNC-TEXT-012 [Post-MVP] Implement spreadsheet function: FIND.
- ☐ SHEETS-FUNC-TEXT-013 [Post-MVP] Implement spreadsheet function: SEARCH.
- ☐ SHEETS-FUNC-TEXT-014 [Post-MVP] Implement spreadsheet function: REPLACE.
- ☐ SHEETS-FUNC-TEXT-015 [Post-MVP] Implement spreadsheet function: SUBSTITUTE.
- ☐ SHEETS-FUNC-TEXT-016 [Post-MVP] Implement spreadsheet function: TEXT.
- ☐ SHEETS-FUNC-TEXT-017 [Post-MVP] Implement spreadsheet function: VALUE.

SHEETS-FUNC-DATE

- ☐ SHEETS-FUNC-DATE-001 [MVP] Implement spreadsheet function: TODAY.
- ☐ SHEETS-FUNC-DATE-002 [MVP] Implement spreadsheet function: NOW.
- ☐ SHEETS-FUNC-DATE-003 [MVP] Implement spreadsheet function: DATE.
- ☐ SHEETS-FUNC-DATE-004 [Post-MVP] Implement spreadsheet function: DATEVALUE.
- ☐ SHEETS-FUNC-DATE-005 [MVP] Implement spreadsheet function: DAY.
- ☐ SHEETS-FUNC-DATE-006 [MVP] Implement spreadsheet function: MONTH.
- ☐ SHEETS-FUNC-DATE-007 [MVP] Implement spreadsheet function: YEAR.
- ☐ SHEETS-FUNC-DATE-008 [Post-MVP] Implement spreadsheet function: EOMONTH.
- ☐ SHEETS-FUNC-DATE-009 [Post-MVP] Implement spreadsheet function: EDATE.
- ☐ SHEETS-FUNC-DATE-010 [Post-MVP] Implement spreadsheet function: WEEKDAY.
- ☐ SHEETS-FUNC-DATE-011 [Post-MVP] Implement spreadsheet function: WEEKNUM.
- ☐ SHEETS-FUNC-DATE-012 [Post-MVP] Implement spreadsheet function: TIME.
- ☐ SHEETS-FUNC-DATE-013 [Post-MVP] Implement spreadsheet function: HOUR.
- ☐ SHEETS-FUNC-DATE-014 [Post-MVP] Implement spreadsheet function: MINUTE.
- ☐ SHEETS-FUNC-DATE-015 [Post-MVP] Implement spreadsheet function: SECOND.
- ☐ SHEETS-FUNC-DATE-016 [Post-MVP] Implement spreadsheet function: NETWORKDAYS.
- ☐ SHEETS-FUNC-DATE-017 [Post-MVP] Implement spreadsheet function: WORKDAY.

SHEETS-FUNC-LOOKUP

- ☐ SHEETS-FUNC-LOOKUP-001 [MVP] Implement spreadsheet function: VLOOKUP.
- ☐ SHEETS-FUNC-LOOKUP-002 [Post-MVP] Implement spreadsheet function: HLOOKUP.
- ☐ SHEETS-FUNC-LOOKUP-003 [Post-MVP] Implement spreadsheet function: XLOOKUP.
- ☐ SHEETS-FUNC-LOOKUP-004 [Post-MVP] Implement spreadsheet function: LOOKUP.
- ☐ SHEETS-FUNC-LOOKUP-005 [Post-MVP] Implement spreadsheet function: INDEX.
- ☐ SHEETS-FUNC-LOOKUP-006 [Post-MVP] Implement spreadsheet function: MATCH.
- ☐ SHEETS-FUNC-LOOKUP-007 [Post-MVP] Implement spreadsheet function: OFFSET.
- ☐ SHEETS-FUNC-LOOKUP-008 [Post-MVP] Implement spreadsheet function: CHOOSE.

SHEETS-FUNC-ARRAY

- ☐ SHEETS-FUNC-ARRAY-001 [Post-MVP] Implement spreadsheet function: TRANSPOSE.
- ☐ SHEETS-FUNC-ARRAY-002 [Post-MVP] Implement spreadsheet function: UNIQUE.
- ☐ SHEETS-FUNC-ARRAY-003 [Post-MVP] Implement spreadsheet function: SORT.
- ☐ SHEETS-FUNC-ARRAY-004 [Post-MVP] Implement spreadsheet function: FILTER.
- ☐ SHEETS-FUNC-ARRAY-005 [Post-MVP] Implement spreadsheet function: SEQUENCE.

SHEETS-FUNC-FIN

- ☐ SHEETS-FUNC-FIN-001 [Post-MVP] Implement spreadsheet function: PMT.
- ☐ SHEETS-FUNC-FIN-002 [Post-MVP] Implement spreadsheet function: FV.
- ☐ SHEETS-FUNC-FIN-003 [Post-MVP] Implement spreadsheet function: PV.
- ☐ SHEETS-FUNC-FIN-004 [Post-MVP] Implement spreadsheet function: NPV.
- ☐ SHEETS-FUNC-FIN-005 [Post-MVP] Implement spreadsheet function: IRR.
- ☐ SHEETS-FUNC-FIN-006 [Post-MVP] Implement spreadsheet function: RATE.

SHEETS-FUNC-INFO

- ☐ SHEETS-FUNC-INFO-001 [MVP] Implement spreadsheet function: ISBLANK.
- ☐ SHEETS-FUNC-INFO-002 [MVP] Implement spreadsheet function: ISNUMBER.
- ☐ SHEETS-FUNC-INFO-003 [MVP] Implement spreadsheet function: ISTEXT.
- ☐ SHEETS-FUNC-INFO-004 [Post-MVP] Implement spreadsheet function: ISLOGICAL.
- ☐ SHEETS-FUNC-INFO-005 [MVP] Implement spreadsheet function: ISERROR.
- ☐ SHEETS-FUNC-INFO-006 [Post-MVP] Implement spreadsheet function: ISERR.
- ☐ SHEETS-FUNC-INFO-007 [MVP] Implement spreadsheet function: ISNA.
- ☐ SHEETS-FUNC-INFO-008 [Post-MVP] Implement spreadsheet function: ISNONTTEXT.
- ☐ SHEETS-FUNC-INFO-009 [Post-MVP] Implement spreadsheet function: ISEVEN.
- ☐ SHEETS-FUNC-INFO-010 [Post-MVP] Implement spreadsheet function: ISODD.
- ☐ SHEETS-FUNC-INFO-011 [Post-MVP] Implement spreadsheet function: ERROR.TYPE.
- ☐ SHEETS-FUNC-INFO-012 [Post-MVP] Implement spreadsheet function: NA.

SHEETS-FUNC-ENG

- ☐ SHEETS-FUNC-ENG-001 [Post-MVP] Implement spreadsheet function: CONVERT.
- ☐ SHEETS-FUNC-ENG-002 [Post-MVP] Implement spreadsheet function: DELTA.
- ☐ SHEETS-FUNC-ENG-003 [Post-MVP] Implement spreadsheet function: ERF.
- ☐ SHEETS-FUNC-ENG-004 [Post-MVP] Implement spreadsheet function: ERFC.
- ☐ SHEETS-FUNC-ENG-005 [Post-MVP] Implement spreadsheet function: GESTEP.
- ☐ SHEETS-FUNC-ENG-006 [Post-MVP] Implement spreadsheet function: HEX2DEC.
- ☐ SHEETS-FUNC-ENG-007 [Post-MVP] Implement spreadsheet function: DEC2HEX.
- ☐ SHEETS-FUNC-ENG-008 [Post-MVP] Implement spreadsheet function: BIN2DEC.
- ☐ SHEETS-FUNC-ENG-009 [Post-MVP] Implement spreadsheet function: DEC2BIN.

SHEETS-FUNC-DB

- ☐ SHEETS-FUNC-DB-001 [Post-MVP] Implement spreadsheet function: DSUM.
- ☐ SHEETS-FUNC-DB-002 [Post-MVP] Implement spreadsheet function: DCOUNT.
- ☐ SHEETS-FUNC-DB-003 [Post-MVP] Implement spreadsheet function: DCOUNTA.
- ☐ SHEETS-FUNC-DB-004 [Post-MVP] Implement spreadsheet function: DAVERAGE.
- ☐ SHEETS-FUNC-DB-005 [Post-MVP] Implement spreadsheet function: DMAX.
- ☐ SHEETS-FUNC-DB-006 [Post-MVP] Implement spreadsheet function: DMIN.
- ☐ SHEETS-FUNC-DB-007 [Post-MVP] Implement spreadsheet function: DGET.

SHEETS-FUNC-WEB

- ☐ SHEETS-FUNC-WEB-001 [Post-MVP] Implement spreadsheet function: WEBSERVICE.

- ☐ SHEETS-FUNC-WEB-002 [Post-MVP] Implement spreadsheet function: ENCODEURL.
- ☐ SHEETS-FUNC-WEB-003 [Post-MVP] Implement spreadsheet function: FILTERXML.

SHEETS-FUNC-SEARCH

- ☐ SHEETS-FUNC-SEARCH-001 [MVP] Implement spreadsheet function: HYPERLINK.
- ☐ SHEETS-FUNC-SEARCH-002 [Post-MVP] Implement spreadsheet function: CELL.
- ☐ SHEETS-FUNC-SEARCH-003 [Post-MVP] Implement spreadsheet function: INFO.
- ☐ SHEETS-FUNC-SEARCH-004 [Post-MVP] Implement spreadsheet function: TYPE.
- ☐ SHEETS-FUNC-SEARCH-005 [Post-MVP] Implement spreadsheet function: N.
- ☐ SHEETS-FUNC-SEARCH-006 [Post-MVP] Implement spreadsheet function: T.

SHEETS-FUNC-TRIG

- ☐ SHEETS-FUNC-TRIG-001 [MVP] Implement spreadsheet function: SIN.
- ☐ SHEETS-FUNC-TRIG-002 [MVP] Implement spreadsheet function: COS.
- ☐ SHEETS-FUNC-TRIG-003 [MVP] Implement spreadsheet function: TAN.
- ☐ SHEETS-FUNC-TRIG-004 [Post-MVP] Implement spreadsheet function: ASIN.
- ☐ SHEETS-FUNC-TRIG-005 [Post-MVP] Implement spreadsheet function: ACOS.
- ☐ SHEETS-FUNC-TRIG-006 [Post-MVP] Implement spreadsheet function: ATAN.
- ☐ SHEETS-FUNC-TRIG-007 [Post-MVP] Implement spreadsheet function: ATAN2.
- ☐ SHEETS-FUNC-TRIG-008 [Post-MVP] Implement spreadsheet function: SINH.
- ☐ SHEETS-FUNC-TRIG-009 [Post-MVP] Implement spreadsheet function: COSH.
- ☐ SHEETS-FUNC-TRIG-010 [Post-MVP] Implement spreadsheet function: TANH.
- ☐ SHEETS-FUNC-TRIG-011 [MVP] Implement spreadsheet function: PI.
- ☐ SHEETS-FUNC-TRIG-012 [Post-MVP] Implement spreadsheet function: RADIANS.
- ☐ SHEETS-FUNC-TRIG-013 [Post-MVP] Implement spreadsheet function: DEGREES.

SHEETS-FUNC-RAND

- ☐ SHEETS-FUNC-RAND-001 [Post-MVP] Implement spreadsheet function: RAND.
- ☐ SHEETS-FUNC-RAND-002 [Post-MVP] Implement spreadsheet function: RANDBETWEEN.

SHEETS-FUNC-MATH2

- ☐ SHEETS-FUNC-MATH2-001 [MVP] Implement spreadsheet function: LOG.
- ☐ SHEETS-FUNC-MATH2-002 [Post-MVP] Implement spreadsheet function: LOG10.
- ☐ SHEETS-FUNC-MATH2-003 [Post-MVP] Implement spreadsheet function: LN.
- ☐ SHEETS-FUNC-MATH2-004 [MVP] Implement spreadsheet function: EXP.
- ☐ SHEETS-FUNC-MATH2-005 [Post-MVP] Implement spreadsheet function: FACT.
- ☐ SHEETS-FUNC-MATH2-006 [Post-MVP] Implement spreadsheet function: FACTDOUBLE.
- ☐ SHEETS-FUNC-MATH2-007 [Post-MVP] Implement spreadsheet function: GCD.
- ☐ SHEETS-FUNC-MATH2-008 [Post-MVP] Implement spreadsheet function: LCM.
- ☐ SHEETS-FUNC-MATH2-009 [Post-MVP] Implement spreadsheet function: COMBIN.
- ☐ SHEETS-FUNC-MATH2-010 [Post-MVP] Implement spreadsheet function: COMBINA.
- ☐ SHEETS-FUNC-MATH2-011 [Post-MVP] Implement spreadsheet function: PERMUT.
- ☐ SHEETS-FUNC-MATH2-012 [Post-MVP] Implement spreadsheet function: PERMUTATIONA.
- ☐ SHEETS-FUNC-MATH2-013 [Post-MVP] Implement spreadsheet function: MROUND.
- ☐ SHEETS-FUNC-MATH2-014 [Post-MVP] Implement spreadsheet function: SIGN.
- ☐ SHEETS-FUNC-MATH2-015 [Post-MVP] Implement spreadsheet function: TRUNC.
- ☐ SHEETS-FUNC-MATH2-016 [Post-MVP] Implement spreadsheet function: RADIANS.
- ☐ SHEETS-FUNC-MATH2-017 [Post-MVP] Implement spreadsheet function: DEGREES.
- ☐ SHEETS-FUNC-MATH2-018 [Post-MVP] Implement spreadsheet function: SERIESSUM.

SHEETS-FUNC-STAT2

- ☐ SHEETS-FUNC-STAT2-001 [Post-MVP] Implement spreadsheet function: CORREL.
- ☐ SHEETS-FUNC-STAT2-002 [Post-MVP] Implement spreadsheet function: COVARIANCE.S.
- ☐ SHEETS-FUNC-STAT2-003 [Post-MVP] Implement spreadsheet function: COVARIANCE.P.
- ☐ SHEETS-FUNC-STAT2-004 [Post-MVP] Implement spreadsheet function: SLOPE.
- ☐ SHEETS-FUNC-STAT2-005 [Post-MVP] Implement spreadsheet function: INTERCEPT.
- ☐ SHEETS-FUNC-STAT2-006 [Post-MVP] Implement spreadsheet function: RSQ.
- ☐ SHEETS-FUNC-STAT2-007 [Post-MVP] Implement spreadsheet function: STEYX.
- ☐ SHEETS-FUNC-STAT2-008 [Post-MVP] Implement spreadsheet function: LINEST.
- ☐ SHEETS-FUNC-STAT2-009 [Post-MVP] Implement spreadsheet function: LOGEST.
- ☐ SHEETS-FUNC-STAT2-010 [Post-MVP] Implement spreadsheet function: FORECAST.LINEAR.
- ☐ SHEETS-FUNC-STAT2-011 [Post-MVP] Implement spreadsheet function: TREND.
- ☐ SHEETS-FUNC-STAT2-012 [Post-MVP] Implement spreadsheet function: GROWTH.
- ☐ SHEETS-FUNC-STAT2-013 [Post-MVP] Implement spreadsheet function: NORM.DIST.
- ☐ SHEETS-FUNC-STAT2-014 [Post-MVP] Implement spreadsheet function: NORM.S.DIST.
- ☐ SHEETS-FUNC-STAT2-015 [Post-MVP] Implement spreadsheet function: NORM.INV.
- ☐ SHEETS-FUNC-STAT2-016 [Post-MVP] Implement spreadsheet function: NORM.S.INV.
- ☐ SHEETS-FUNC-STAT2-017 [Post-MVP] Implement spreadsheet function: T.DIST.
- ☐ SHEETS-FUNC-STAT2-018 [Post-MVP] Implement spreadsheet function: T.INV.
- ☐ SHEETS-FUNC-STAT2-019 [Post-MVP] Implement spreadsheet function: CHISQ.DIST.
- ☐ SHEETS-FUNC-STAT2-020 [Post-MVP] Implement spreadsheet function: CHISQ.INV.
- ☐ SHEETS-FUNC-STAT2-021 [Post-MVP] Implement spreadsheet function: F.DIST.
- ☐ SHEETS-FUNC-STAT2-022 [Post-MVP] Implement spreadsheet function: F.INV.
- ☐ SHEETS-FUNC-STAT2-023 [Post-MVP] Implement spreadsheet function: BINOM.DIST.
- ☐ SHEETS-FUNC-STAT2-024 [Post-MVP] Implement spreadsheet function: POISSON.DIST.
- ☐ SHEETS-FUNC-STAT2-025 [Post-MVP] Implement spreadsheet function: EXPON.DIST.

SHEETS-FUNC-TEXT2

- ☐ SHEETS-FUNC-TEXT2-001 [Post-MVP] Implement spreadsheet function: EXACT.
- ☐ SHEETS-FUNC-TEXT2-002 [Post-MVP] Implement spreadsheet function: REPT.
- ☐ SHEETS-FUNC-TEXT2-003 [Post-MVP] Implement spreadsheet function: CHAR.
- ☐ SHEETS-FUNC-TEXT2-004 [Post-MVP] Implement spreadsheet function: CODE.
- ☐ SHEETS-FUNC-TEXT2-005 [Post-MVP] Implement spreadsheet function: UNICODE.
- ☐ SHEETS-FUNC-TEXT2-006 [Post-MVP] Implement spreadsheet function: UNICHAR.
- ☐ SHEETS-FUNC-TEXT2-007 [Post-MVP] Implement spreadsheet function: TEXTAFTER.
- ☐ SHEETS-FUNC-TEXT2-008 [Post-MVP] Implement spreadsheet function: TEXTBEFORE.
- ☐ SHEETS-FUNC-TEXT2-009 [Post-MVP] Implement spreadsheet function: TEXTSPLIT.
- ☐ SHEETS-FUNC-TEXT2-010 [Post-MVP] Implement spreadsheet function: TEXTJOIN.
- ☐ SHEETS-FUNC-TEXT2-011 [Post-MVP] Implement spreadsheet function: SORTBY.
- ☐ SHEETS-FUNC-TEXT2-012 [Post-MVP] Implement spreadsheet function: CLEAN.
- ☐ SHEETS-FUNC-TEXT2-013 [Post-MVP] Implement spreadsheet function: PROPER.
- ☐ SHEETS-FUNC-TEXT2-014 [Post-MVP] Implement spreadsheet function: DOLLAR.
- ☐ SHEETS-FUNC-TEXT2-015 [Post-MVP] Implement spreadsheet function: FIXED.

SHEETS-FUNC-DATE2

- ☐ SHEETS-FUNC-DATE2-001 [Post-MVP] Implement spreadsheet function: DATEDIF.
- ☐ SHEETS-FUNC-DATE2-002 [Post-MVP] Implement spreadsheet function: DAYS.
- ☐ SHEETS-FUNC-DATE2-003 [Post-MVP] Implement spreadsheet function: DAYS360.
- ☐ SHEETS-FUNC-DATE2-004 [Post-MVP] Implement spreadsheet function: YEARFRAC.
- ☐ SHEETS-FUNC-DATE2-005 [Post-MVP] Implement spreadsheet function: TIMEVALUE.
- ☐ SHEETS-FUNC-DATE2-006 [MVP] Implement spreadsheet function: HOUR.
- ☐ SHEETS-FUNC-DATE2-007 [MVP] Implement spreadsheet function: MINUTE.

- ☐ SHEETS-FUNC-DATE2-008 [MVP] Implement spreadsheet function: SECOND.
- ☐ SHEETS-FUNC-DATE2-009 [Post-MVP] Implement spreadsheet function: WEEKDAY.
- ☐ SHEETS-FUNC-DATE2-010 [Post-MVP] Implement spreadsheet function: WEEKNUM.
- ☐ SHEETS-FUNC-DATE2-011 [Post-MVP] Implement spreadsheet function: WORKDAY.
- ☐ SHEETS-FUNC-DATE2-012 [Post-MVP] Implement spreadsheet function: WORKDAY.INTL.
- ☐ SHEETS-FUNC-DATE2-013 [Post-MVP] Implement spreadsheet function: NETWORKDAYS.
- ☐ SHEETS-FUNC-DATE2-014 [Post-MVP] Implement spreadsheet function: NETWORKDAYS.INTL.

SHEETS-FUNC-LOOKUP2

- ☐ SHEETS-FUNC-LOOKUP2-001 [Post-MVP] Implement spreadsheet function: XLOOKUP.
- ☐ SHEETS-FUNC-LOOKUP2-002 [Post-MVP] Implement spreadsheet function: XMATCH.
- ☐ SHEETS-FUNC-LOOKUP2-003 [MVP] Implement spreadsheet function: MATCH.
- ☐ SHEETS-FUNC-LOOKUP2-004 [MVP] Implement spreadsheet function: INDEX.
- ☐ SHEETS-FUNC-LOOKUP2-005 [Post-MVP] Implement spreadsheet function: INDIRECT.
- ☐ SHEETS-FUNC-LOOKUP2-006 [Post-MVP] Implement spreadsheet function: ADDRESS.
- ☐ SHEETS-FUNC-LOOKUP2-007 [MVP] Implement spreadsheet function: ROW.
- ☐ SHEETS-FUNC-LOOKUP2-008 [MVP] Implement spreadsheet function: ROWS.
- ☐ SHEETS-FUNC-LOOKUP2-009 [MVP] Implement spreadsheet function: COLUMN.
- ☐ SHEETS-FUNC-LOOKUP2-010 [MVP] Implement spreadsheet function: COLUMNS.
- ☐ SHEETS-FUNC-LOOKUP2-011 [Post-MVP] Implement spreadsheet function: HYPERLINK.

SHEETS-FUNC-ARRAY2

- ☐ SHEETS-FUNC-ARRAY2-001 [Post-MVP] Implement spreadsheet function: LET.
- ☐ SHEETS-FUNC-ARRAY2-002 [Post-MVP] Implement spreadsheet function: LAMBDA.
- ☐ SHEETS-FUNC-ARRAY2-003 [Post-MVP] Implement spreadsheet function: MAP.
- ☐ SHEETS-FUNC-ARRAY2-004 [Post-MVP] Implement spreadsheet function: REDUCE.
- ☐ SHEETS-FUNC-ARRAY2-005 [Post-MVP] Implement spreadsheet function: SCAN.
- ☐ SHEETS-FUNC-ARRAY2-006 [Post-MVP] Implement spreadsheet function: BYROW.
- ☐ SHEETS-FUNC-ARRAY2-007 [Post-MVP] Implement spreadsheet function: BYCOL.
- ☐ SHEETS-FUNC-ARRAY2-008 [Post-MVP] Implement spreadsheet function: TAKE.
- ☐ SHEETS-FUNC-ARRAY2-009 [Post-MVP] Implement spreadsheet function: DROP.
- ☐ SHEETS-FUNC-ARRAY2-010 [Post-MVP] Implement spreadsheet function: TOCOL.
- ☐ SHEETS-FUNC-ARRAY2-011 [Post-MVP] Implement spreadsheet function: TOROW.
- ☐ SHEETS-FUNC-ARRAY2-012 [Post-MVP] Implement spreadsheet function: HSTACK.
- ☐ SHEETS-FUNC-ARRAY2-013 [Post-MVP] Implement spreadsheet function: VSTACK.
- ☐ SHEETS-FUNC-ARRAY2-014 [Post-MVP] Implement spreadsheet function: WRAPROWS.
- ☐ SHEETS-FUNC-ARRAY2-015 [Post-MVP] Implement spreadsheet function: WRAPCOLS.

SHEETS-FUNC-FIN2

- ☐ SHEETS-FUNC-FIN2-001 [Post-MVP] Implement spreadsheet function: IPMT.
- ☐ SHEETS-FUNC-FIN2-002 [Post-MVP] Implement spreadsheet function: PPMT.
- ☐ SHEETS-FUNC-FIN2-003 [Post-MVP] Implement spreadsheet function: CUMIPMT.
- ☐ SHEETS-FUNC-FIN2-004 [Post-MVP] Implement spreadsheet function: CUMPRINC.
- ☐ SHEETS-FUNC-FIN2-005 [Post-MVP] Implement spreadsheet function: XNPV.
- ☐ SHEETS-FUNC-FIN2-006 [Post-MVP] Implement spreadsheet function: XIRR.
- ☐ SHEETS-FUNC-FIN2-007 [Post-MVP] Implement spreadsheet function: SLN.
- ☐ SHEETS-FUNC-FIN2-008 [Post-MVP] Implement spreadsheet function: SYD.
- ☐ SHEETS-FUNC-FIN2-009 [Post-MVP] Implement spreadsheet function: DB.
- ☐ SHEETS-FUNC-FIN2-010 [Post-MVP] Implement spreadsheet function: DDB.

SHEETS-TEST

- ☐ SHEETS-TEST-001 [MVP] Integration tests for SHEETS-GRID: selection, edit, clipboard, undo/redo, persistence.
- ☐ SHEETS-TEST-002 [MVP] Performance benchmarks for SHEETS-GRID: large sheets, scroll, calc time.
- ☐ SHEETS-TEST-003 [MVP] Integration tests for SHEETS-CALC: selection, edit, clipboard, undo/redo, persistence.
- ☐ SHEETS-TEST-004 [MVP] Performance benchmarks for SHEETS-CALC: large sheets, scroll, calc time.
- ☐ SHEETS-TEST-005 [MVP] Integration tests for SHEETS-FORMAT: selection, edit, clipboard, undo/redo, persistence.
- ☐ SHEETS-TEST-006 [MVP] Performance benchmarks for SHEETS-FORMAT: large sheets, scroll, calc time.
- ☐ SHEETS-TEST-007 [MVP] Integration tests for SHEETS-DATA: selection, edit, clipboard, undo/redo, persistence.
- ☐ SHEETS-TEST-008 [MVP] Performance benchmarks for SHEETS-DATA: large sheets, scroll, calc time.
- ☐ SHEETS-TEST-009 [MVP] Integration tests for SHEETS-CHART: selection, edit, clipboard, undo/redo, persistence.
- ☐ SHEETS-TEST-010 [MVP] Performance benchmarks for SHEETS-CHART: large sheets, scroll, calc time.

FORMATS-XLSX

- ☐ FORMATS-XLSX-001 [Post-MVP] XLSX import: sheets + cells + basic formulas.
- ☐ FORMATS-XLSX-002 [Post-MVP] XLSX import: formats (basic).
- ☐ FORMATS-XLSX-003 [Post-MVP] XLSX export: sheets + cells + formulas.
- ☐ FORMATS-XLSX-004 [Post-MVP] XLSX export: formats (basic).
- ☐ FORMATS-XLSX-005 [Post-MVP] XLSX export: charts (post-MVP).

Slides Tasks

SLIDES-CORE

- ☐ SLIDES-CORE-001 [MVP] Slide thumbnail pane with drag-drop reorder and multi-select.
- ☐ SLIDES-CORE-002 [MVP] Canvas renderer with zoom and pan.
- ☐ SLIDES-CORE-003 [MVP] Object selection: click, marquee, multi-select; keyboard nudge.
- ☐ SLIDES-CORE-004 [MVP] Undo/redo for object edits and slide operations.
- ☐ SLIDES-CORE-005 [MVP] Slide layouts MVP: title, title+content, two-column, blank; apply/reset.
- ☐ SLIDES-CORE-006 [MVP] Themes MVP: apply preset theme (colors + fonts).
- ☐ SLIDES-CORE-007 [MVP] Slide notes editor per slide.

SLIDES-TEXT

- ☐ SLIDES-TEXT-001 [MVP] Text box object: create, edit, move, resize.
- ☐ SLIDES-TEXT-002 [MVP] Typography: font, size, bold/italic/underline, color, alignment.
- ☐ SLIDES-TEXT-003 [MVP] Bullets/numbering (basic).
- ☐ SLIDES-TEXT-004 [Post-MVP] Auto-fit text .

SLIDES-SHAPES

- ☐ SLIDES-SHAPES-001 [MVP] Basic shapes: rectangle, ellipse, line, arrow.
- ☐ SLIDES-SHAPES-002 [MVP] Fill/stroke settings; opacity (basic).
- ☐ SLIDES-SHAPES-003 [MVP] Arrange: bring forward/back, align/distribute, group/ungroup.
- ☐ SLIDES-SHAPES-004 [MVP] Snap guides + smart guides; optional grid.

SLIDES-MEDIA

- ☐ SLIDES-MEDIA-001 [MVP] Insert images: file picker/paste/drag; resize/crop (basic).
- ☐ SLIDES-MEDIA-002 [MVP] Alt text for images.
- ☐ SLIDES-MEDIA-003 [Post-MVP] Embed video .

SLIDES-MOTION

- ☐ SLIDES-MOTION-001 [MVP] Transitions MVP: none, fade, push, wipe; duration control.
- ☐ SLIDES-MOTION-002 [MVP] Animations MVP: appear/fade; per-object order list (basic).
- ☐ SLIDES-MOTION-003 [Post-MVP] Post-MVP: advanced animations, motion paths, triggers, morph-like transitions.

SLIDES-SHOW

- ☐ SLIDES-SHOW-001 [MVP] Slide show full-screen playback with keyboard navigation.
- ☐ SLIDES-SHOW-002 [MVP] Presenter view MVP: current slide + notes (single screen) (limited).
- ☐ SLIDES-SHOW-003 [MVP] Export to PDF MVP: render slides and build PDF.
- ☐ SLIDES-SHOW-004 [Post-MVP] Export to images (PNG) .

SLIDES-TRANSITIONS

- ☐ SLIDES-TRANSITIONS-001 [Post-MVP] Support slide transition: Cut.
- ☐ SLIDES-TRANSITIONS-002 [MVP] Support slide transition: Fade.
- ☐ SLIDES-TRANSITIONS-003 [MVP] Support slide transition: Push.
- ☐ SLIDES-TRANSITIONS-004 [MVP] Support slide transition: Wipe.
- ☐ SLIDES-TRANSITIONS-005 [Post-MVP] Support slide transition: Split.
- ☐ SLIDES-TRANSITIONS-006 [Post-MVP] Support slide transition: Reveal.
- ☐ SLIDES-TRANSITIONS-007 [Post-MVP] Support slide transition: Random Bars.
- ☐ SLIDES-TRANSITIONS-008 [Post-MVP] Support slide transition: Shape.
- ☐ SLIDES-TRANSITIONS-009 [Post-MVP] Support slide transition: Uncover.

- ☐ SLIDES-TRANSITIONS-010 [Post-MVP] Support slide transition: Cover.
- ☐ SLIDES-TRANSITIONS-011 [Post-MVP] Support slide transition: Flash.
- ☐ SLIDES-TRANSITIONS-012 [Post-MVP] Support slide transition: Morph.

SLIDES-ANIM

- ☐ SLIDES-ANIM-001 [MVP] Support object animation: Appear.
- ☐ SLIDES-ANIM-002 [MVP] Support object animation: Fade.
- ☐ SLIDES-ANIM-003 [Post-MVP] Support object animation: Fly In.
- ☐ SLIDES-ANIM-004 [Post-MVP] Support object animation: Float In.
- ☐ SLIDES-ANIM-005 [Post-MVP] Support object animation: Split.
- ☐ SLIDES-ANIM-006 [Post-MVP] Support object animation: Wipe.
- ☐ SLIDES-ANIM-007 [Post-MVP] Support object animation: Zoom.
- ☐ SLIDES-ANIM-008 [Post-MVP] Support object animation: Bounce.
- ☐ SLIDES-ANIM-009 [Post-MVP] Support object animation: Grow/Shrink.
- ☐ SLIDES-ANIM-010 [Post-MVP] Support object animation: Spin.
- ☐ SLIDES-ANIM-011 [Post-MVP] Support object animation: Motion Path: Lines.
- ☐ SLIDES-ANIM-012 [Post-MVP] Support object animation: Motion Path: Arcs.

SLIDES-SHAPES-CAT

- ☐ SLIDES-SHAPES-CAT-001 [MVP] Support shape primitive: Rectangle.
- ☐ SLIDES-SHAPES-CAT-002 [Post-MVP] Support shape primitive: Rounded Rectangle.
- ☐ SLIDES-SHAPES-CAT-003 [MVP] Support shape primitive: Ellipse.
- ☐ SLIDES-SHAPES-CAT-004 [MVP] Support shape primitive: Triangle.
- ☐ SLIDES-SHAPES-CAT-005 [Post-MVP] Support shape primitive: Right Triangle.
- ☐ SLIDES-SHAPES-CAT-006 [MVP] Support shape primitive: Diamond.
- ☐ SLIDES-SHAPES-CAT-007 [Post-MVP] Support shape primitive: Parallelogram.
- ☐ SLIDES-SHAPES-CAT-008 [Post-MVP] Support shape primitive: Trapezoid.
- ☐ SLIDES-SHAPES-CAT-009 [Post-MVP] Support shape primitive: Pentagon.
- ☐ SLIDES-SHAPES-CAT-010 [Post-MVP] Support shape primitive: Hexagon.
- ☐ SLIDES-SHAPES-CAT-011 [Post-MVP] Support shape primitive: Heptagon.
- ☐ SLIDES-SHAPES-CAT-012 [Post-MVP] Support shape primitive: Octagon.
- ☐ SLIDES-SHAPES-CAT-013 [Post-MVP] Support shape primitive: Star (5).
- ☐ SLIDES-SHAPES-CAT-014 [Post-MVP] Support shape primitive: Star (8).
- ☐ SLIDES-SHAPES-CAT-015 [Post-MVP] Support shape primitive: Callout.
- ☐ SLIDES-SHAPES-CAT-016 [MVP] Support shape primitive: Arrow (right).
- ☐ SLIDES-SHAPES-CAT-017 [Post-MVP] Support shape primitive: Arrow (left).
- ☐ SLIDES-SHAPES-CAT-018 [Post-MVP] Support shape primitive: Arrow (up).
- ☐ SLIDES-SHAPES-CAT-019 [Post-MVP] Support shape primitive: Arrow (down).
- ☐ SLIDES-SHAPES-CAT-020 [MVP] Support shape primitive: Line.
- ☐ SLIDES-SHAPES-CAT-021 [Post-MVP] Support shape primitive: Connector (elbow).
- ☐ SLIDES-SHAPES-CAT-022 [Post-MVP] Support shape primitive: Connector (straight).

SLIDES-LAYOUTS

- ☐ SLIDES-LAYOUTS-001 [MVP] Provide slide layout template: Title Slide.
- ☐ SLIDES-LAYOUTS-002 [MVP] Provide slide layout template: Title and Content.
- ☐ SLIDES-LAYOUTS-003 [Post-MVP] Provide slide layout template: Section Header.
- ☐ SLIDES-LAYOUTS-004 [MVP] Provide slide layout template: Two Content.
- ☐ SLIDES-LAYOUTS-005 [Post-MVP] Provide slide layout template: Comparison.
- ☐ SLIDES-LAYOUTS-006 [MVP] Provide slide layout template: Title Only.
- ☐ SLIDES-LAYOUTS-007 [MVP] Provide slide layout template: Blank.
- ☐ SLIDES-LAYOUTS-008 [Post-MVP] Provide slide layout template: Content with Caption.
- ☐ SLIDES-LAYOUTS-009 [Post-MVP] Provide slide layout template: Picture with Caption.

SLIDES-RIBBON

- ☐ SLIDES-RIBBON-001 [MVP] Slides ribbon: Home -> New Slide.
- ☐ SLIDES-RIBBON-002 [MVP] Slides ribbon: Home -> Layout.
- ☐ SLIDES-RIBBON-003 [MVP] Slides ribbon: Home -> Reset.
- ☐ SLIDES-RIBBON-004 [Post-MVP] Slides ribbon: Home -> Section.
- ☐ SLIDES-RIBBON-005 [MVP] Slides ribbon: Home -> Font controls.
- ☐ SLIDES-RIBBON-006 [MVP] Slides ribbon: Home -> Paragraph controls.
- ☐ SLIDES-RIBBON-007 [MVP] Slides ribbon: Home -> Drawing tools.
- ☐ SLIDES-RIBBON-008 [MVP] Slides ribbon: Home -> Arrange (align/distribute).
- ☐ SLIDES-RIBBON-009 [MVP] Slides ribbon: Insert -> Text box.
- ☐ SLIDES-RIBBON-010 [MVP] Slides ribbon: Insert -> Pictures.
- ☐ SLIDES-RIBBON-011 [MVP] Slides ribbon: Insert -> Shapes.
- ☐ SLIDES-RIBBON-012 [Post-MVP] Slides ribbon: Insert -> Icons.
- ☐ SLIDES-RIBBON-013 [Post-MVP] Slides ribbon: Insert -> Chart.
- ☐ SLIDES-RIBBON-014 [Post-MVP] Slides ribbon: Insert -> Table.
- ☐ SLIDES-RIBBON-015 [Post-MVP] Slides ribbon: Insert -> Audio.
- ☐ SLIDES-RIBBON-016 [Post-MVP] Slides ribbon: Insert -> Video.
- ☐ SLIDES-RIBBON-017 [Post-MVP] Slides ribbon: Insert -> Link.
- ☐ SLIDES-RIBBON-018 [MVP] Slides ribbon: Design -> Themes gallery.
- ☐ SLIDES-RIBBON-019 [Post-MVP] Slides ribbon: Design -> Variants.
- ☐ SLIDES-RIBBON-020 [Post-MVP] Slides ribbon: Design -> Slide size.
- ☐ SLIDES-RIBBON-021 [Post-MVP] Slides ribbon: Design -> Format background.
- ☐ SLIDES-RIBBON-022 [MVP] Slides ribbon: Transitions -> Transition gallery.
- ☐ SLIDES-RIBBON-023 [Post-MVP] Slides ribbon: Transitions -> Effect options.
- ☐ SLIDES-RIBBON-024 [MVP] Slides ribbon: Transitions -> Duration.
- ☐ SLIDES-RIBBON-025 [Post-MVP] Slides ribbon: Transitions -> Apply to all.
- ☐ SLIDES-RIBBON-026 [MVP] Slides ribbon: Animations -> Animation gallery.
- ☐ SLIDES-RIBBON-027 [Post-MVP] Slides ribbon: Animations -> Animation pane.
- ☐ SLIDES-RIBBON-028 [Post-MVP] Slides ribbon: Animations -> Trigger/timing.
- ☐ SLIDES-RIBBON-029 [MVP] Slides ribbon: Slide Show -> From beginning/current slide.
- ☐ SLIDES-RIBBON-030 [MVP] Slides ribbon: Slide Show -> Presenter view.
- ☐ SLIDES-RIBBON-031 [Post-MVP] Slides ribbon: Slide Show -> Set up show.
- ☐ SLIDES-RIBBON-032 [Post-MVP] Slides ribbon: Review -> Comments.
- ☐ SLIDES-RIBBON-033 [Post-MVP] Slides ribbon: Review -> Spelling.
- ☐ SLIDES-RIBBON-034 [Post-MVP] Slides ribbon: View -> Normal/Slide sorter/Notes page.
- ☐ SLIDES-RIBBON-035 [Post-MVP] Slides ribbon: View -> Guides/Gridlines.
- ☐ SLIDES-RIBBON-036 [Post-MVP] Slides ribbon: View -> Slide Master view.
- ☐ SLIDES-RIBBON-037 [MVP] Slides ribbon: View -> Zoom.

SLIDES-TEST

- ☐ SLIDES-TEST-001 [MVP] Integration tests for SLIDES-CORE: object manipulation, undo/redo, persistence.
- ☐ SLIDES-TEST-002 [MVP] Export validation tests for SLIDES-CORE: PDF output and visual regression checks.
- ☐ SLIDES-TEST-003 [MVP] Integration tests for SLIDES-TEXT: object manipulation, undo/redo, persistence.
- ☐ SLIDES-TEST-004 [MVP] Export validation tests for SLIDES-TEXT: PDF output and visual regression checks.
- ☐ SLIDES-TEST-005 [MVP] Integration tests for SLIDES-SHAPES: object manipulation, undo/redo, persistence.
- ☐ SLIDES-TEST-006 [MVP] Export validation tests for SLIDES-SHAPES: PDF output and visual regression checks.
- ☐ SLIDES-TEST-007 [MVP] Integration tests for SLIDES-MEDIA: object manipulation, undo/redo, persistence.
- ☐ SLIDES-TEST-008 [MVP] Export validation tests for SLIDES-MEDIA: PDF output and visual regression checks.
- ☐ SLIDES-TEST-009 [MVP] Integration tests for SLIDES-MOTION: object manipulation, undo/redo, persistence.
- ☐ SLIDES-TEST-010 [MVP] Export validation tests for SLIDES-MOTION: PDF output and visual regression checks.
- ☐ SLIDES-TEST-011 [MVP] Integration tests for SLIDES-SHOW: object manipulation, undo/redo, persistence.

- ☐ SLIDES-TEST-012 [MVP] Export validation tests for SLIDES-SHOW: PDF output and visual regression checks.

FORMATS-PPTX

- ☐ FORMATS-PPTX-001 [Post-MVP] PPTX import: slides + text boxes.
- ☐ FORMATS-PPTX-002 [Post-MVP] PPTX import: images.
- ☐ FORMATS-PPTX-003 [Post-MVP] PPTX import: themes (basic).
- ☐ FORMATS-PPTX-004 [Post-MVP] PPTX export: slides + text boxes.
- ☐ FORMATS-PPTX-005 [Post-MVP] PPTX export: images.
- ☐ FORMATS-PPTX-006 [Post-MVP] PPTX export: themes (basic).

Base Tasks

BASE-NAV

- ☐ BASE-NAV-001 [MVP] Left navigation tree: Tables, Queries, Forms, Reports.
- ☐ BASE-NAV-002 [MVP] Tabbed documents area for designers and datasheets.
- ☐ BASE-NAV-003 [MVP] Inspector pane for field properties and validation rules.
- ☐ BASE-NAV-004 [MVP] Mobile wizard flows for table/field creation.

BASE-TABLES

- ☐ BASE-TABLES-001 [MVP] Create/edit/delete table; rename table.
- ☐ BASE-TABLES-002 [MVP] Field designer: add/edit/delete fields; field types (text, number, date, boolean, autonumber, choice).
- ☐ BASE-TABLES-003 [MVP] Primary key support; unique constraint support (basic).
- ☐ BASE-TABLES-004 [MVP] Datasheet virtualization: scroll through large tables (pagination).
- ☐ BASE-TABLES-005 [MVP] Row editing: add/delete, inline validation errors, undo/redo.
- ☐ BASE-TABLES-006 [MVP] Sort and filter columns in datasheet.
- ☐ BASE-TABLES-007 [MVP] CSV import/export per table.

BASE-REL

- ☐ BASE-REL-001 [MVP] Foreign key fields and one-to-many relationships.
- ☐ BASE-REL-002 [MVP] Referential integrity checks (restrict) on delete/update.
- ☐ BASE-REL-003 [Post-MVP] Lookup field UI: show related record label (post-MVP advanced).

BASE-QUERY

- ☐ BASE-QUERY-001 [MVP] Query builder MVP: select fields, simple filters, sort.
- ☐ BASE-QUERY-002 [MVP] Join builder MVP: choose tables and join keys (inner join).
- ☐ BASE-QUERY-003 [MVP] Saved queries: store builder state + generated SQL/AST.
- ☐ BASE-QUERY-004 [MVP] Query runner results grid with export CSV.

BASE-QUERY-FEAT

- ☐ BASE-QUERY-FEAT-001 [Post-MVP] Query builder: calculated fields (expressions).
- ☐ BASE-QUERY-FEAT-002 [Post-MVP] Query builder: aggregates (COUNT/SUM/AVG) and GROUP BY.
- ☐ BASE-QUERY-FEAT-003 [Post-MVP] Parameterized queries with prompts.
- ☐ BASE-QUERY-FEAT-004 [Post-MVP] Saved query dependencies and refresh behavior.
- ☐ BASE-QUERY-FEAT-005 [Post-MVP] Expression functions: CONCAT, LEFT, RIGHT, DATEADD (subset).

BASE-FIELD-PROPS

- ☐ BASE-FIELD-PROPS-001 [MVP] Support Base field property: Default value.
- ☐ BASE-FIELD-PROPS-002 [MVP] Support Base field property: Validation rule.
- ☐ BASE-FIELD-PROPS-003 [MVP] Support Base field property: Validation text.
- ☐ BASE-FIELD-PROPS-004 [Post-MVP] Support Base field property: Input mask.
- ☐ BASE-FIELD-PROPS-005 [Post-MVP] Support Base field property: Format.
- ☐ BASE-FIELD-PROPS-006 [Post-MVP] Support Base field property: Caption.
- ☐ BASE-FIELD-PROPS-007 [MVP] Support Base field property: Required.
- ☐ BASE-FIELD-PROPS-008 [Post-MVP] Support Base field property: Allow zero length.
- ☐ BASE-FIELD-PROPS-009 [MVP] Support Base field property: Indexed (No/Duplicates OK/No Duplicates).
- ☐ BASE-FIELD-PROPS-010 [MVP] Support Base field property: Lookup (value list).
- ☐ BASE-FIELD-PROPS-011 [Post-MVP] Support Base field property: Lookup (table relationship).
- ☐ BASE-FIELD-PROPS-012 [Post-MVP] Support Base field property: Attachment (post-MVP).

BASE-FORMS

- ☐ BASE-FORMS-001 [MVP] Auto-generate form from table/query.
- ☐ BASE-FORMS-002 [MVP] Form controls: text input, number, date picker, checkbox, dropdown for choices.
- ☐ BASE-FORMS-003 [MVP] Validation messages and required field indicators.
- ☐ BASE-FORMS-004 [MVP] Form navigation: next/prev record, new record, delete record.

BASE-FORMS-CONTROLS

- ☐ BASE-FORMS-CONTROLS-001 [MVP] Form designer: support control type: Text box.
- ☐ BASE-FORMS-CONTROLS-002 [MVP] Form designer: support control type: Number input.
- ☐ BASE-FORMS-CONTROLS-003 [MVP] Form designer: support control type: Date picker.
- ☐ BASE-FORMS-CONTROLS-004 [MVP] Form designer: support control type: Checkbox.
- ☐ BASE-FORMS-CONTROLS-005 [MVP] Form designer: support control type: Dropdown (choice).
- ☐ BASE-FORMS-CONTROLS-006 [Post-MVP] Form designer: support control type: Lookup dropdown.
- ☐ BASE-FORMS-CONTROLS-007 [MVP] Form designer: support control type: Textarea (multi-line).
- ☐ BASE-FORMS-CONTROLS-008 [Post-MVP] Form designer: support control type: Radio group.
- ☐ BASE-FORMS-CONTROLS-009 [Post-MVP] Form designer: support control type: File attachment.
- ☐ BASE-FORMS-CONTROLS-010 [Post-MVP] Form designer: support control type: Image control.

BASE-REPORTS

- ☐ BASE-REPORTS-001 [MVP] Auto-generate list report from query.
- ☐ BASE-REPORTS-002 [MVP] Grouping by one field (basic) + group headers.
- ☐ BASE-REPORTS-003 [MVP] Print/PDF export for reports (basic).

BASE-REPORTS-FEAT

- ☐ BASE-REPORTS-FEAT-001 [MVP] Report designer feature: Page header/footer.
- ☐ BASE-REPORTS-FEAT-002 [MVP] Report designer feature: Page numbers.
- ☐ BASE-REPORTS-FEAT-003 [MVP] Report designer feature: Grouping levels (multi).
- ☐ BASE-REPORTS-FEAT-004 [MVP] Report designer feature: Sorting.
- ☐ BASE-REPORTS-FEAT-005 [Post-MVP] Report designer feature: Conditional formatting.
- ☐ BASE-REPORTS-FEAT-006 [Post-MVP] Report designer feature: Calculated controls.
- ☐ BASE-REPORTS-FEAT-007 [Post-MVP] Report designer feature: Charts on reports.
- ☐ BASE-REPORTS-FEAT-008 [Post-MVP] Report designer feature: Subreports.

BASE-RIBBON

- ☐ BASE-RIBBON-001 [MVP] Base ribbon: Create -> Create table.
- ☐ BASE-RIBBON-002 [MVP] Base ribbon: Create -> Create query.
- ☐ BASE-RIBBON-003 [MVP] Base ribbon: Create -> Create form.
- ☐ BASE-RIBBON-004 [MVP] Base ribbon: Create -> Create report.
- ☐ BASE-RIBBON-005 [MVP] Base ribbon: External Data -> Import CSV.
- ☐ BASE-RIBBON-006 [MVP] Base ribbon: External Data -> Export CSV.
- ☐ BASE-RIBBON-007 [Post-MVP] Base ribbon: External Data -> Link external database.
- ☐ BASE-RIBBON-008 [MVP] Base ribbon: Database Tools -> Relationships view.
- ☐ BASE-RIBBON-009 [Post-MVP] Base ribbon: Database Tools -> Database documenter.
- ☐ BASE-RIBBON-010 [Post-MVP] Base ribbon: Database Tools -> Compact/repair.
- ☐ BASE-RIBBON-011 [MVP] Base ribbon: Fields -> Add field.
- ☐ BASE-RIBBON-012 [MVP] Base ribbon: Fields -> Data type picker.
- ☐ BASE-RIBBON-013 [MVP] Base ribbon: Fields -> Validation.
- ☐ BASE-RIBBON-014 [MVP] Base ribbon: Fields -> Indexes.
- ☐ BASE-RIBBON-015 [MVP] Base ribbon: Table Design -> Primary key.
- ☐ BASE-RIBBON-016 [MVP] Base ribbon: Table Design -> Property sheet.
- ☐ BASE-RIBBON-017 [MVP] Base ribbon: Table Design -> Field order.

BASE-TEST

- ☐ BASE-TEST-001 [MVP] Integration tests for BASE-NAV: schema edits, row edits, queries, persistence.
- ☐ BASE-TEST-002 [MVP] Data integrity tests for BASE-NAV: constraints, relationships, export/import.
- ☐ BASE-TEST-003 [MVP] Integration tests for BASE-TABLES: schema edits, row edits, queries, persistence.
- ☐ BASE-TEST-004 [MVP] Data integrity tests for BASE-TABLES: constraints, relationships, export/import.
- ☐ BASE-TEST-005 [MVP] Integration tests for BASE-REL: schema edits, row edits, queries, persistence.
- ☐ BASE-TEST-006 [MVP] Data integrity tests for BASE-REL: constraints, relationships, export/import.
- ☐ BASE-TEST-007 [MVP] Integration tests for BASE-QUERY: schema edits, row edits, queries, persistence.
- ☐ BASE-TEST-008 [MVP] Data integrity tests for BASE-QUERY: constraints, relationships, export/import.
- ☐ BASE-TEST-009 [MVP] Integration tests for BASE-FORMS: schema edits, row edits, queries, persistence.
- ☐ BASE-TEST-010 [MVP] Data integrity tests for BASE-FORMS: constraints, relationships, export/import.
- ☐ BASE-TEST-011 [MVP] Integration tests for BASE-REPORTS: schema edits, row edits, queries, persistence.
- ☐ BASE-TEST-012 [MVP] Data integrity tests for BASE-REPORTS: constraints, relationships, export/import.